

Development of Automated Administrative Services Management System for the Department of CSE, IUBAT

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ID# 21203010

A Practicum in the Partial Fulfillment of the Requirements
for the Award of Bachelor of Computer Science and Engineering (BCSE)



Department of Computer Science and Engineering
College of Engineering and Technology
IUBAT–International University of Business Agriculture and Technology

Spring 2025

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The practicum has been examined and approved,

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Department of Computer Science and Engineering
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Spring 2025

Letter of Transmittal

08 February 2025

The Chair

Practicum Defense Committee

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4 Embankment Drive Road, Sector 10, Uttara Model Town

Dhaka 1230, Bangladesh.

Subject: Letter of Transmittal.

Dear Sir,

With due respect, I would like to inform you that it is a great pleasure for me to submit this report entitled **“Development of Automated Administrative Services Management System for the Department of CSE, IUBAT”** for the fulfillment of my practicum course.

It was a great opportunity for me to work on this project to actualize my theoretical knowledge more realistic and I gained a lot of exposure with the corporate culture of renowned company. Now I am looking forward to your kind commentary on this practicum report.

I shall always be very grateful to you if you kindly go through this report and evaluate my performance

Yours sincerely,

Shahidul Alam
ID# 21203010

Organization Certificate

Student's Declaration

I hereby declare that this practicum report titled Development of Automated Administrative Services Management System for the Department of CSE, IUBAT is uniquely prepared by me with team lead instruction during the completion of a three-month internship at the Department of CSE, IUBAT. I also confirm that the report is only prepared for the partial requirement of my internship and not for any other purpose, reward, or degree at IUBAT or any other institution either by me or any other student. I also declare that there is no data falsification and materials used in this report from various sources have been duly cited.

Shahidul Alam
ID# 21203010

Supervisor's Certification

This certificate confirms that the Practicum report titled Development of Automated Administrative Services Management System for the Department of CSE, IUBAT^{II} has been conducted by Shahidul Alam identified by ID# 21203010 at IUBAT- International University of Business Agriculture and Technology as part of the requirements for the practicum defense course. This report was created under my supervision and represents a comprehensive record of successfully completed work. As far as my knowledge and the provided declaration are concerned, no portions of this report have been previously submitted for any degree, diploma, or certification. Shahidul Alam is now authorized to submit the report, and I extend my best wishes for his success in all his future endeavors

Engr A.S.M Shakil Ahamed
Supervisor , Senior Lecturer
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Abstract

The Automated Administrative Services Management System is a web-based application designed to streamline administrative tasks and enhance student satisfaction within the Department of Computer Science and Engineering at IUBAT. This system automates various manual processes, reducing physical labor for the administrative staff and minimizing waiting times for students. It provides efficient service management, user-friendly interfaces, and secure data handling. Key features include user management, course management, semester management, timetable management, request handling, pre-course offering, and report generation. The system is built using an incremental software process model, ensuring progressive development and early delivery of essential functionalities. This project aims to demonstrate the potential of modern technology in improving administrative efficiency and overall student experience. Future enhancements could include integrating more advanced features and expanding the system's capabilities.

Acknowledgments

In the name of ALLAH, who is the most merciful and the most graceful. This report for Automated Administrative Services Management System^{ll} would not have been possible without the essential and gracious support of many individuals who encouraged me to complete this project on time. I want to express my gratitude & respect to my honorable Coordinator Md. Rashedul Islam & Chairman Prof Dr Utpal Kanti Das, Dept. of Computer Science and Engineering, IUBAT– International University of Business Agriculture and Technology for their constant guidance, encouragement & every possible help in the overall preparation of this report. I shall always remain thankful for the advice and suggestions provided by my Supervisor, Engr A.S.M Shakil Ahamed, Department of Computer Science and Engineering, IUBAT– International University of Business Agriculture and Technology. I would also like to thank all of my friends for their valuable suggestions and comments. Finally, I wish to thank my parent who has been a great source of inspiration to me. Without their support, I would not have reached where I am today.

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Chapter 1

Introduction

An internship gives you real-world experience that you can use in your future job. It can also be seen as a test run for any company to help you get ready for work life and start building your career. This is the first step in explaining the project's goal, scope, limits, and everything else that comes up during the start-up phase.

1.1 Project Overview

The project —Development of Automated Administrative Services Management System for the Department of CSE, IUBATl is designed to streamline the administrative processes for students and staff in the department. This software is developed to reduce physical labor, minimize waiting times, and enhance the overall service management experience for students and administrative staff.

The system includes comprehensive user and system requirements ensuring a user-friendly experience and robust functionality. It provides features such as practicum and thesis management, log-book management, semester management, timetable management, routine management, room allocation, and section generation, enhancing efficiency and communication.

1.2 Objectives

The ultimate objective of the Automated Administrative Services Management System is to provide an efficient experience to students and administrators by streamlining administrative processes. The project aims to deliver a sophisticated and reliable web application that meets the needs of both students and department administrators.

1.2.1 Broad Objectives

The broad objectives aim to create a robust and adaptable Automated Administrative Services Management System that enhances student satisfaction, facilitates efficient department management, and improves the overall educational experience.

1.2.2 Specific Objectives

- **Secure Login and User Management:** Ensure that users can securely log in to the system and manage their accounts effectively.

- **Pre-Practicum and Pre-Thesis Offering:** Give Notice and manage pre-practicum/Thesis offering and student participation.
- **Notice Management:** Can manage notices by ensuring timely creation, distribution, and tracking of updates, thereby enhancing communication, maintaining compliance.
- **Manage Log-Books:** Students can perform given task and Teacher can assign task to students.
- **Manage Routine:** Can generate routine easily and view course routine and teacher routine.
- **Thesis and Practicum Supervisor Assignment:** Assign supervisors to students for thesis and practicum.
- **Event Management:** Create, edit, and view departmental events. Event insights with charts (registration, participation, etc.).
- **Counseling Management:** Teachers can create and manage counseling schedules. Students can request counseling by selecting faculty, date, and time
- **Request Management:** Students can submit various service requests.
- **Statistics & Insights:** Visual dashboards for events, requests, counseling, and more

1.3 Scope of the Project

The scope of the project is focused on delivering a fully functional and secure Automated Administrative Services Management System that meets the needs of students and administrators while ensuring data accuracy and system reliability. The system is designed to streamline the course enrollment process, making it easier for students to manage their courses and for administrators to handle requests and scheduling.

1.4 Limitation of the Project

The hands-on experience during the practicum at the department has been enlightening,

bridging the gap between theoretical knowledge and real-world application. However, the duration of 12-14 weeks seems a bit short for a recent graduate like me to fully grasp and adapt to the practical aspects. My project involves a real-life course management system with complexities such as managing multiple course requests, generating routines, and user-based functionality serving a large number of students and administrators daily across the department. The challenge lies in optimizing the software for such a large scale. Implementing thorough software engineering procedures has been difficult within the tight timeframe, but with time, I've gained a better understanding of the overall situation.

1.5 Methodology

The plan for creating the "Automated Administrative Services Management System" involves following specific steps outlined in Software Analysis & Design. This study focuses on developing a web-based app to meet the needs of over 2000 students of the Department of Computer Science & Engineering. We'll figure out the important factors to control by analyzing data from various sources, both primary and secondary.

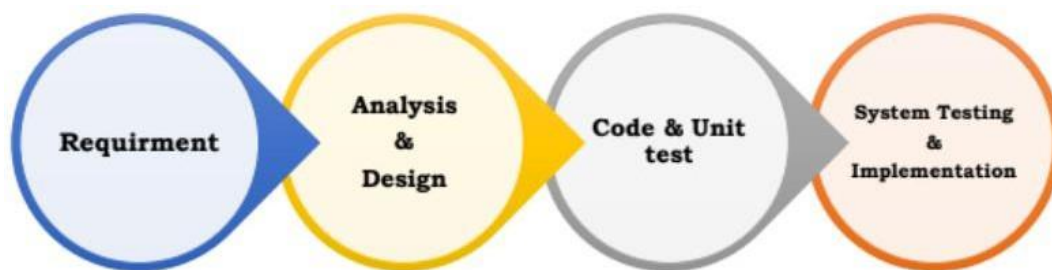


Figure 1.1 Software Life Cycles

1.6 Software Process Model

In some situations, the initial software plans make sense, but the development process faces challenges due to its extensive scope. Software development often follows a progressive model where the product is conceptualized, used, built, and tested in evolving stages. Many businesses and system providers prefer this incremental software development approach. It's suitable for projects where the software requirements are clear but the actual implementation may take some time, and there's an early need for essential software functions.

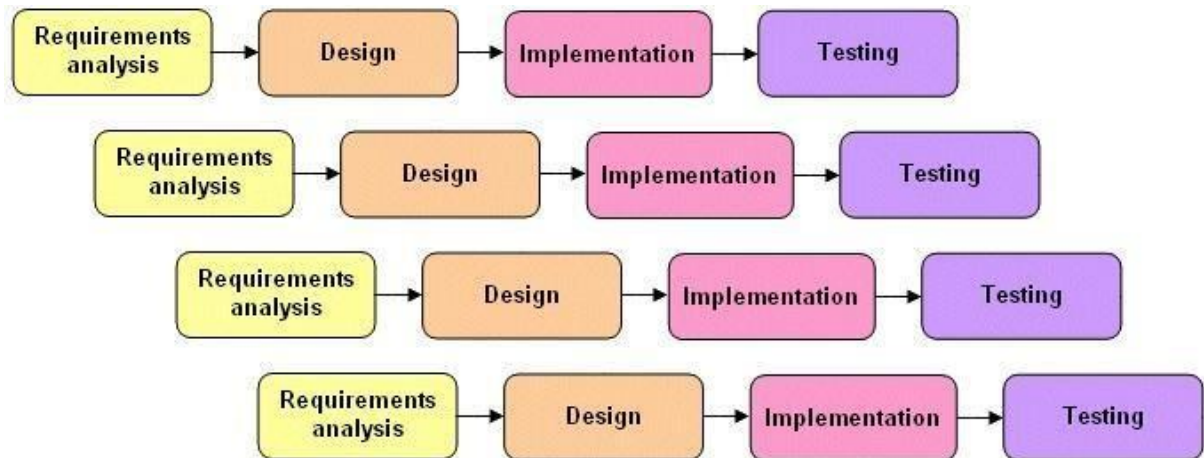


Figure 1.2 Incremental Process Model

1.7 Incremental Model Steps

- **Make a partial version of a full system:** Begin by developing a functional subset of the entire system.
- **Add more features one by one:** Gradually integrate additional features in subsequent iterations.
- **System requirements are prioritized and implemented in groups:** Organize and address requirements in manageable increments.
- **System is updated with each new version:** Continuously enhance the system with each development cycle until all planned functionalities are incorporated.

1.8 Strength

- **Early Software Delivery:** Provides functional software early in the life cycle.
- **Flexibility:** Accommodates changes in requirements with less expense.
- **Improved Testing and Debugging:** Simplifies testing and debugging with fewer iterations.
- **Risk Management:** Identifies and manages risks more effectively during iterations.

- **Significant Progress:** Demonstrates substantial progress at each iteration.
- **Early Access to Key Features:** Delivers important functionalities to users early.
- **Reduced Risk of Requirement Changes:** Mitigates risks associated with evolving requirements.

When to use the Incremental model?

- **All system requirements are known and understood:** Suitable when complete requirements are available.
- **Anticipated Requirement Changes:** Ideal when initial requirements are known, but likely to change over time.
- **Early Feature Delivery:** Beneficial when essential features need to be delivered quickly.
- **Resource Constraints:** Useful when resources with the necessary skill set are limited.
- **High-Risk Characteristics and Goals:** Applicable for projects with certain high-risk attributes and objectives

Benefits by using incremental model

The main benefit of using this model is that it helps speed up system development. In the incremental model, I start by creating important features and quickly share them with the user for testing. After that, I work on adding more functions. This method lets everyone involved in the project see actual progress, making it clear that things are moving forward. Following the incremental model's principles also brings additional advantages such as:

- **Enhanced Flexibility:** Easier to accommodate changes in requirements.
- **Improved Risk Management:** Facilitates early identification and mitigation of risks.
- **Better Testing and Debugging:** Simplifies the process by focusing on smaller, manageable increments.
- **Continuous Progress:** Provides visible progress throughout the development cycle.

- **Disadvantages of Incremental model**
- **Requires Careful Planning and Layout:** Demands meticulous planning and design upfront.
- **Comprehensive Definition Needed:** Requires a thorough definition of the system before breaking it down into increments.

1.9 Feasibility Studies

Feasibility analysis checks if a proposed solution is practical and worthwhile for the company. It helps decide if the tasks are worth doing. When looking into a new system, three main areas are explored to generate ideas. These three key factors are considered to assess the system's feasibility, deciding if it's suitable for automation.

- Technical feasibility.
- Economic feasibility.
- Operational feasibility.

Technical Feasibility: This study examines the hardware and software needed for the proposed system to carry out its processes.

Table 1.1 Software and Hardware Requirements

Hardware Requirements	Software Requirements
Computer (Laptop, Desktop), Server.	OS (Windows)
	Browser (Edge, Chrome)
Fulltime electricity support.	React, Dot net core Web API
4GB system memory with 1 TB HDD.	MySQL, VS Code, Visual Studio

Communication Interface

1. Client on Internet will be using HTTP/HTTPS protocol.
2. Client on Internet will be using TCP/IP protocol.

Economic feasibility

Economic feasibility assesses the cost-effectiveness of implementing a new system. It involves evaluating whether the company can financially support the redesign and whether the project itself will yield a cost-effective outcome.

Operational feasibility

Operational feasibility encompasses considerations related to user acceptance, management endorsement, and alignment with the needs of various entities and external factors. The new system aims to mitigate most of the shortcomings of the current system and is expected to gain acceptance from both parties.

Chapter 2

Organizational Overview

The Department of Computer Science and Engineering at IUBAT is a dynamic and fast-moving department that has proven itself to be one of the high performers in the field of computer science and engineering. Established in 1991, we have students from all over the world.

2.1 Overview of Department of Computer Science and Engineering, IUBAT

At the Department of Computer Science and Engineering, we've achieved great success by turning challenges into opportunities and transforming obstacles into research prospects. Our commitment to serving students has made us integral partners in their achievements. We have some outstanding stories that truly stand out and deserve recognition.

2.2 We are promised to help Society

We believe that numerous organizations encounter situations similar to the ones we've successfully improved. We've helped tackle challenges such as overlooked research opportunities, not using resources to the fullest, and students not fully utilizing system capabilities. For these common issues, we have ready-made solutions at your service. Additionally, if you encounter any unique challenges, our department is ready to help you find customized solutions.

2.3 Organization's Philosophy

We follow all regulatory regulations as well as any requests from our students to the letter. We constantly put our students at the center of everything we do. Some of our compliance certifications include, but are not limited to, the ones listed below.

2.4 Services

The Department of Computer Science and Engineering at IUBAT provides a range of services to its faculty and students, including but not limited to:

- Educational Services: The department offers a comprehensive curriculum covering various areas of computer science and engineering. This includes classroom instruction, laboratory work, and project-based learning.

- **Research Opportunities:** Faculty and students in the department engage in cutting-edge research in various areas of computer science and engineering. The department provides the necessary support and resources for such research activities.
- **Internship Opportunities:** The department can offer internships to students. These internships provide practical experience, helping students apply the knowledge they've gained in the classroom to real-world problems.
- **System/Software Development:** The department has the capability to build systems or software to help students and aid in the management of the department. This could include learning management systems, administrative tools, and more.

2.5 Organization Location

ADDRESS: IUBAT—International University of Business Agriculture and Technology 4 Embankment Drive Road, Sector 10, Uttara Model Town, Dhaka 1230, Bangladesh

2.6 Vision

Our vision incorporates the following:

- With a strong dedication to quality and student satisfaction in the computer science and engineering industry.
- To be regarded as one of the most admired computer science and engineering departments, adhering to and supporting environmentally responsible processes at all stages of education.
- Dealing with all students with a high level of ethics and fairness.

2.7 Mission

Our mission is to be impactful by bringing value to our students. We want to leverage our students' education and research implementations through:

- Independent and trusted advice
- Custom solutions built on nouveau technologies
- Superior utilization of our skilled professionals

2.8 Organizational Structure

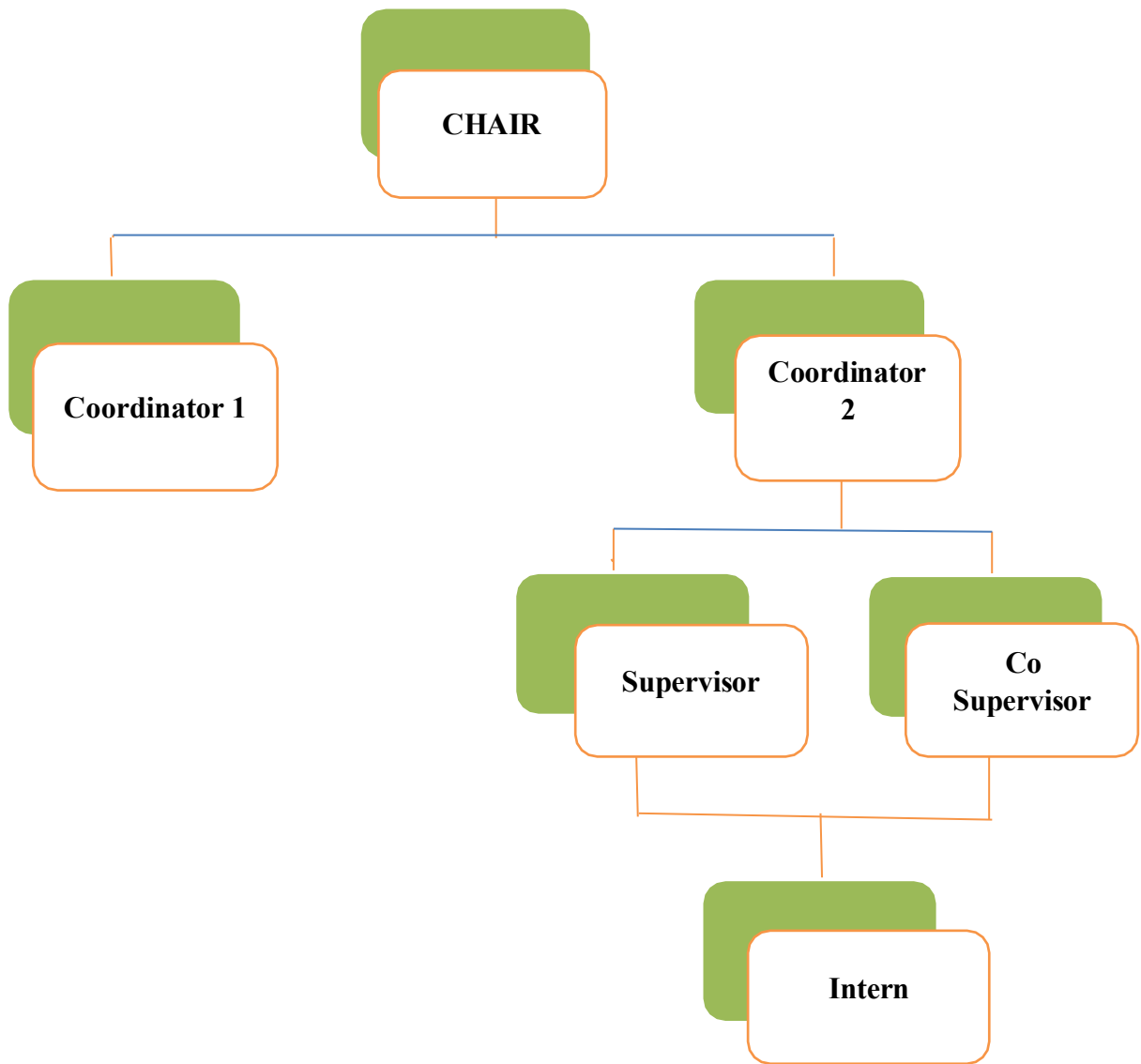


Figure 2.1 Organizational Structure

Chapter 3

Requirement Engineering

This paper describes a methodology for the generation of software requirements for the Automated Administrative Services Management System at IUBAT.

3.1 Requirements Engineering

Requirements engineering is a branch of engineering that focuses on specifying software systems and the needs of end users. It is all about understanding what users want from a computer system and how those wants affect the architecture of the system. Software engineering focuses on building the solution that consumer's desire.

1. User Requirement
2. System Requirement
3. Functional Requirement
4. Non-Functional Requirement

3.1.1 User Requirement

AO Requirements:

1. **User Management:**
 - Add, update, and manage teachers and students.
 - Manage user authentication and access control.
2. **Pre-Practicum Management:**
 - Batch-wise practicum offering.
 - Add, delete, update pre-practicum.
3. **Pre-Thesis Management:**
 - Batch-wise practicum offering.
 - Add, delete, update pre-thesis.
4. **Timetable Management:**
 - Create, update, view timetables.

5. Notice Management:

- Create, update, delete, view notice.
- Handle notice chair/coordinator.

Coordinator Requirements:

6. Room Allocation:

- Add, Update, Delete Room.

7. Section Allocation:

- Can generate section list by given input of student number.

8. Teacher Information Management:

- Add Teacher information.

9. Generate Routine:

- Can generate weekly routine for course and teacher.

10. View Routine:

- Can view course and teacher routine.

Teacher Requirements:

1. Routine View:

- Access and review assigned routines.

2. Access Log Book:

- Assign task to students in log book.
- Update task status.

3. View Notice:

- Access and view notices.

4. Communication:

- Communicate with students regarding tasks.

Student Requirements:

1. Login:

- Secure authentication.

2. View Notice:

- View all notices and access links.

3. Perform Pre-Offering:

- Fill-up pre-offering form.

4. Log Book:

- Perform given task by supervisor.
- View task status.

3.1.2 System Requirement

Admin Functions:

1. Secure login and user management.
2. Notice, Log-Book and Routine management.
3. Timetable management.
4. Pre-Practicum offering application.
5. Pre-Thesis offering application.
6. Generate and View Routine.

Teacher Functions:

1. Secure login and routine review.

2. Assign task and update status log-book management.
3. View routine and notices.

Student Functions:

1. Secure login and course view/enrollment.
2. Perform log-book task and check status.
3. View notices

3.1.3 Functional Requirement

- **User Management:** Create, update, and delete user accounts. Ensure secure authentication and authorization.
- **Pre-Practicum and Pre-Thesis Offering:** Add, update, and delete Offering.
- **Timetable Management:** Create and manage timetables.
- **Notice Management:** Post, update and delete notices based on the situation and importance.
- **Generate Routine:** Create, update, delete and view course and teacher routine.

3.1.4 Non-Functional Requirement

- **Secure User Authentication System:** Ensure robust security protocols for user authentication and data protection.
- **Intuitive and User-Friendly Interface:** Design an interface that is easy to navigate and use.
- **Data Integrity and Reliability:** Maintain accuracy and reliability of all data processed and stored.
- **System Performance Optimization:** Ensure the system performs efficiently and responds promptly to user actions.

- **Scalability:** Design the system to accommodate a growing user base and expanding course offerings. Sure, I can help refine your use case descriptions. Here's an improved version:
- **Timely Response to User Actions:** Minimize delays in system response times to user requests.

3.2 Use Case Diagram

In a use case, some important parts include:

- **Use Case:** Illustrated as a rounded shape containing the user's task description, signifying a specific interaction or behavior within the modeled system.
- **Actor:** Depicted as a stick figure or a simple rectangle with the actor's name inscribed, representing an external entity engaging with the modeled system.
- **System Boundary:** Displayed as a rectangle, or a rectangle with rounded corners, encompassing both use cases and actors, outlining the system's scope.
- **Association:** The representation is indicated by a solid line connecting an actor to a use case, or linking two use cases, signifying a relationship between them.
- **Include:** Visualized as a dashed line with an arrowhead pointing to the included use case, indicating an inclusive relationship.
- **Extend:** Portrayed as a dashed line with a pointed arrowhead, symbolizing an extension connection between use cases. This signifies that one use case's action can be expanded or augmented by another under specific conditions.

Use Case symbols:

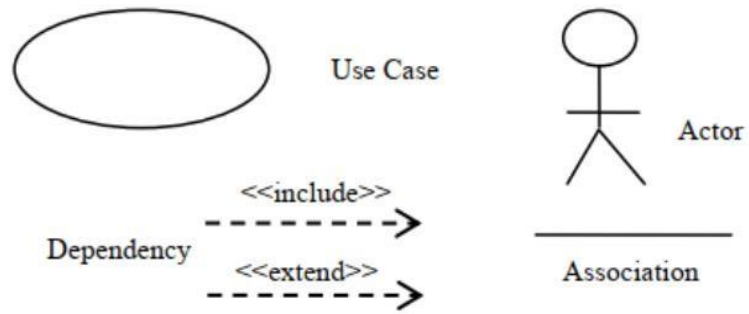


Figure 3.1: Use Case Symbols

Diagram Representation:

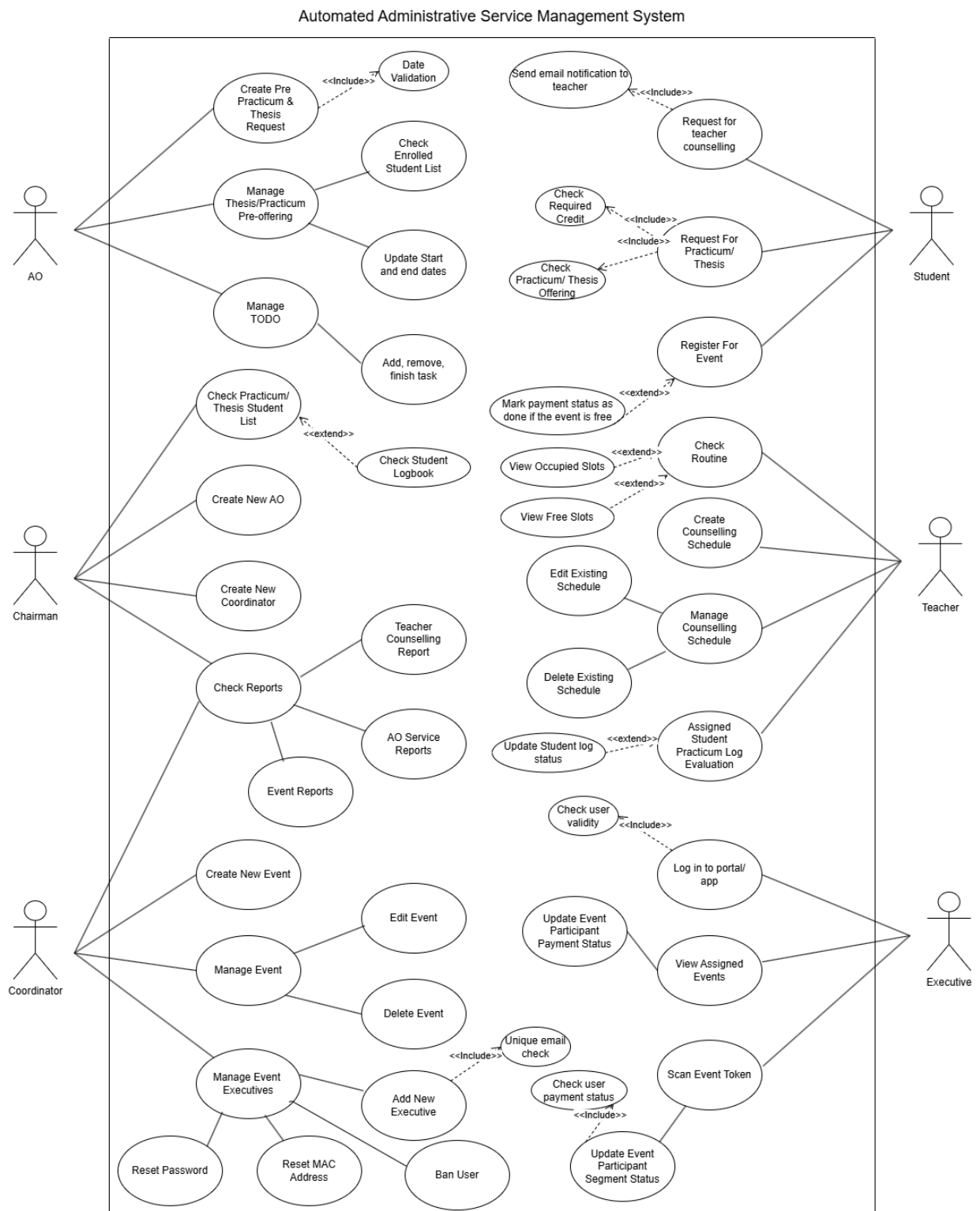


Figure 3.2: Use Case Diagram

Chapter 4

Analysis Modeling

To facilitate effective communication of data, functions, and behavioral requirements, analysis modeling combines clear textual descriptions with illustrative diagrams. This combination not only improves understanding but also enables a detailed examination for accuracy, comprehensiveness, and coherence. In this realm, available resources include traditional approaches, object-oriented analysis (OOA), and Unified Modeling Language (UML). Particularly, the 4.1 Activity Diagram plays a crucial role in explaining these complex aspects.

The goals of analysis modeling include –

- Examination of a domain for comprehensive understanding.
- Precise specification of client requirements.
- Formation of a robust framework guiding software design development.
- Compilation of a checklist of specifications for post-development software validation.

4.1 Software Analysis Pattern

Conceptual models, referred to as software analysis patterns, encapsulate abstractions of frequently encountered scenarios in modeling. Often termed analysis patterns in software engineering, they consist of a set of interconnected generic objects (meta-classes) defined by precise attributes (data definitions), behaviors (method signatures), and expected interactions. These analysis patterns provide a basis for the model creator to effectively address system requirements.

It is imperative-

- Define client expectations clearly.
- Enumerate verifiable requirements for ongoing software development.
- Establish a framework for shaping the software design process

4.2 Activity Diagram

Activity diagrams in UML serve as pivotal behavioral illustrations, portraying the intricate dynamics within a system. Resembling sophisticated flow charts, these diagrams adeptly capture the seamless transition between diverse activities. They intricately showcase

the chronological sequence of activities, offering a comprehensive depiction of the evolving states throughout the process.

4.2.1 Activity Diagram for Teacher Counselling Request

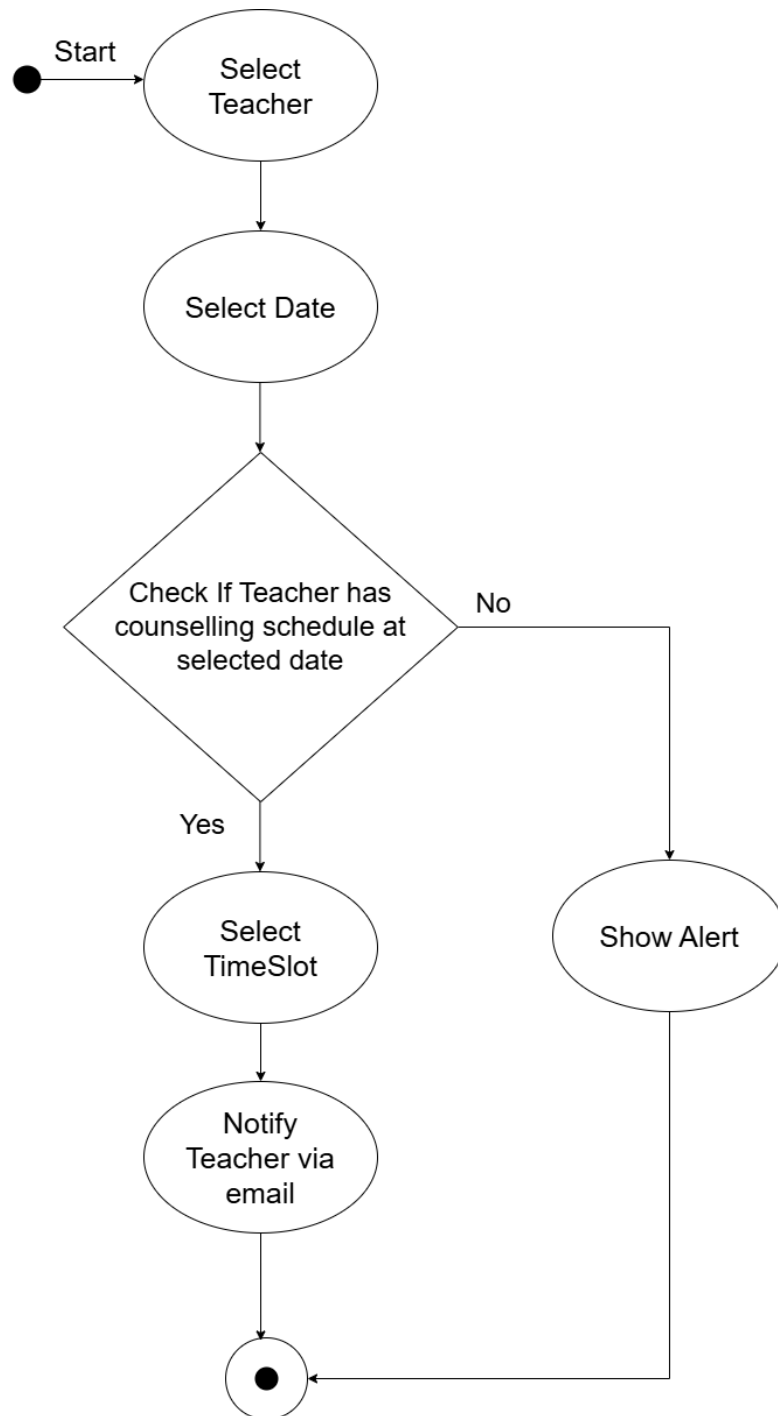


Figure 4.1: Activity Diagram for Teacher Counselling Request

4.2.2 Activity Diagram Student Practicum Request

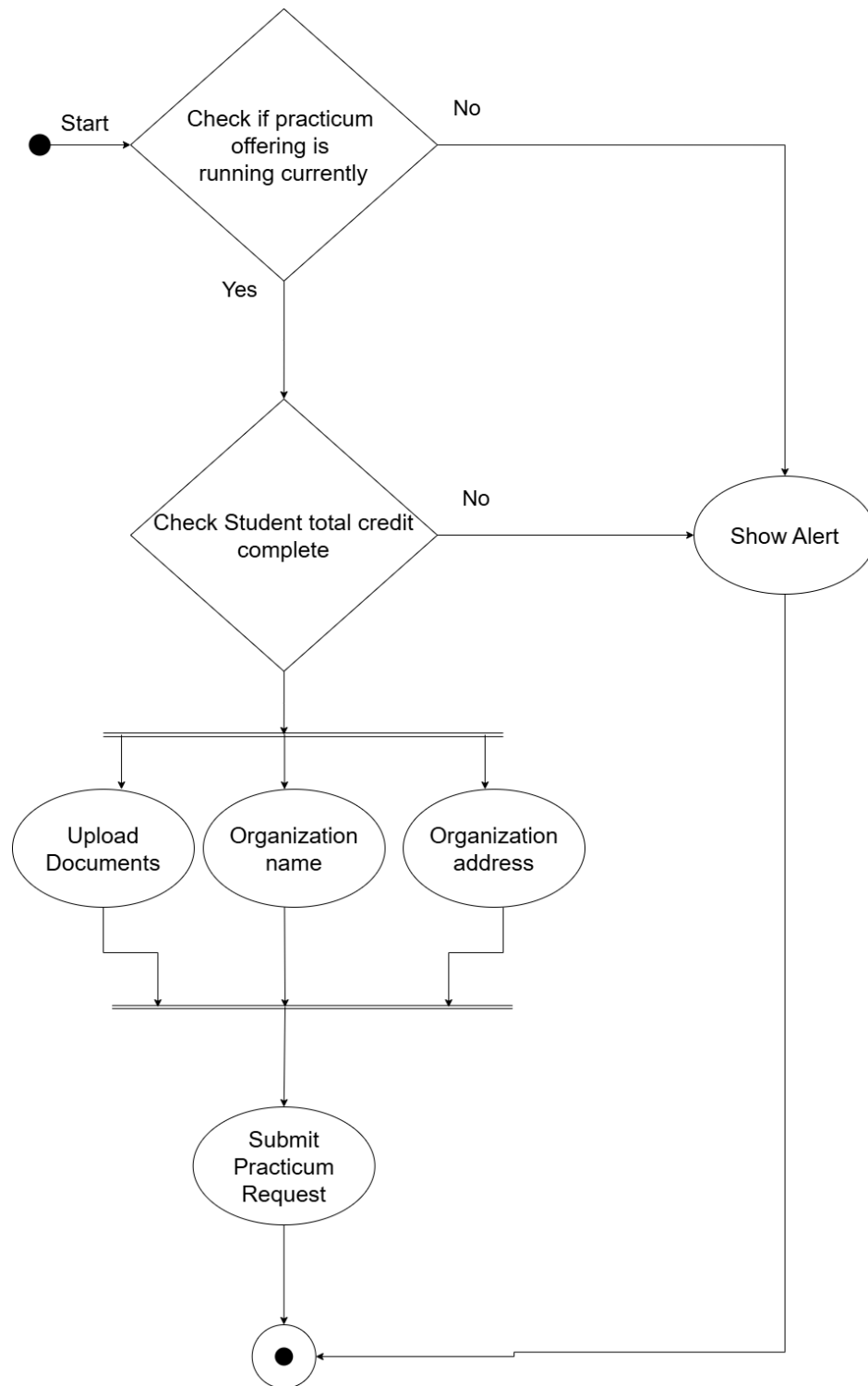


Figure 4.2: Activity Diagram for Student Practicum Request

4.2.3 Activity Diagram for Event Registration

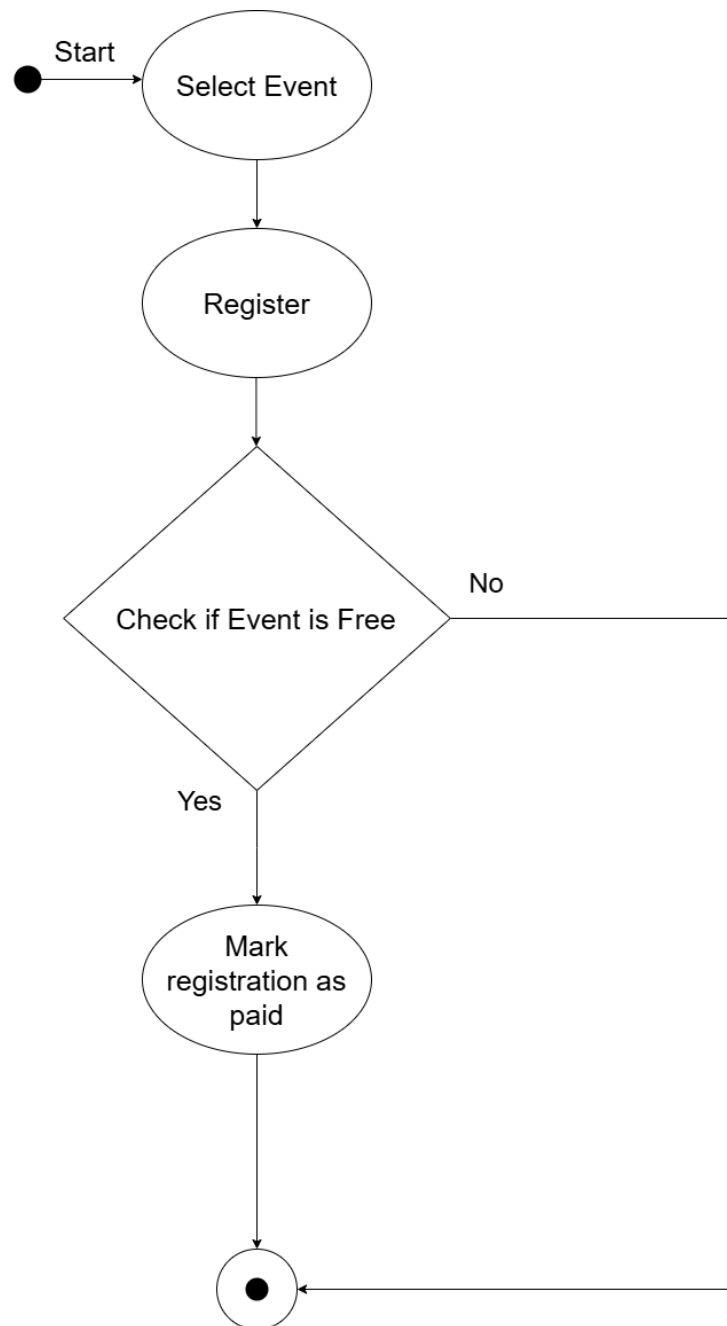


Figure 4.3: Activity Diagram for Event Registration

4.2.4 Activity Diagram for Creating Teacher Counselling Schedule

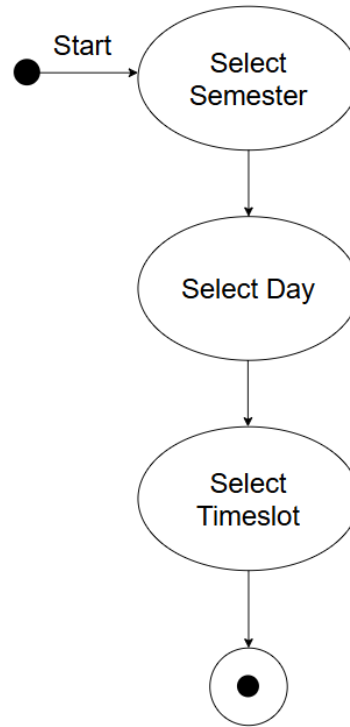


Figure 4.4: Activity Diagram for Creating Teacher Counselling Schedule

4.2.5 Activity Diagram for Executive Mobile App Login

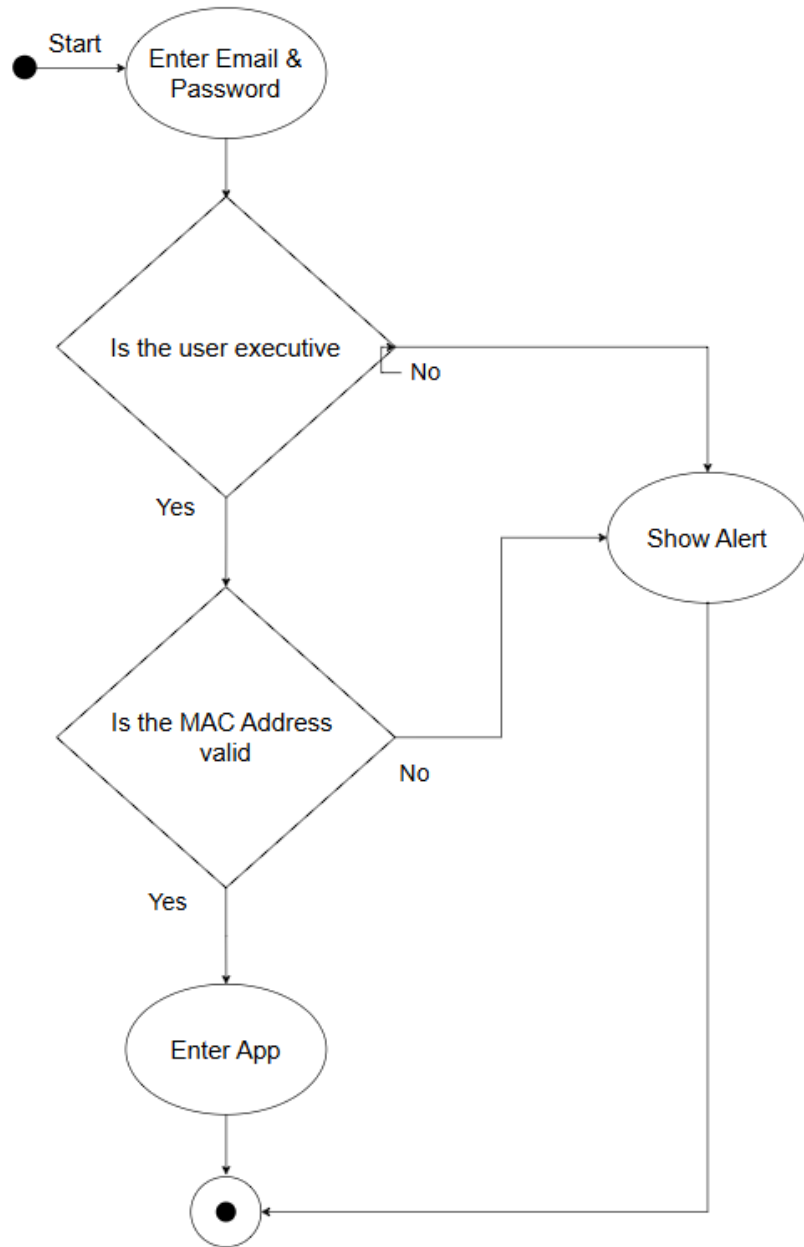


Figure 4.5: Activity Diagram for Executive Mobile App Login

4.2.6 Activity Diagram for Toggle Event Participant Payment Status

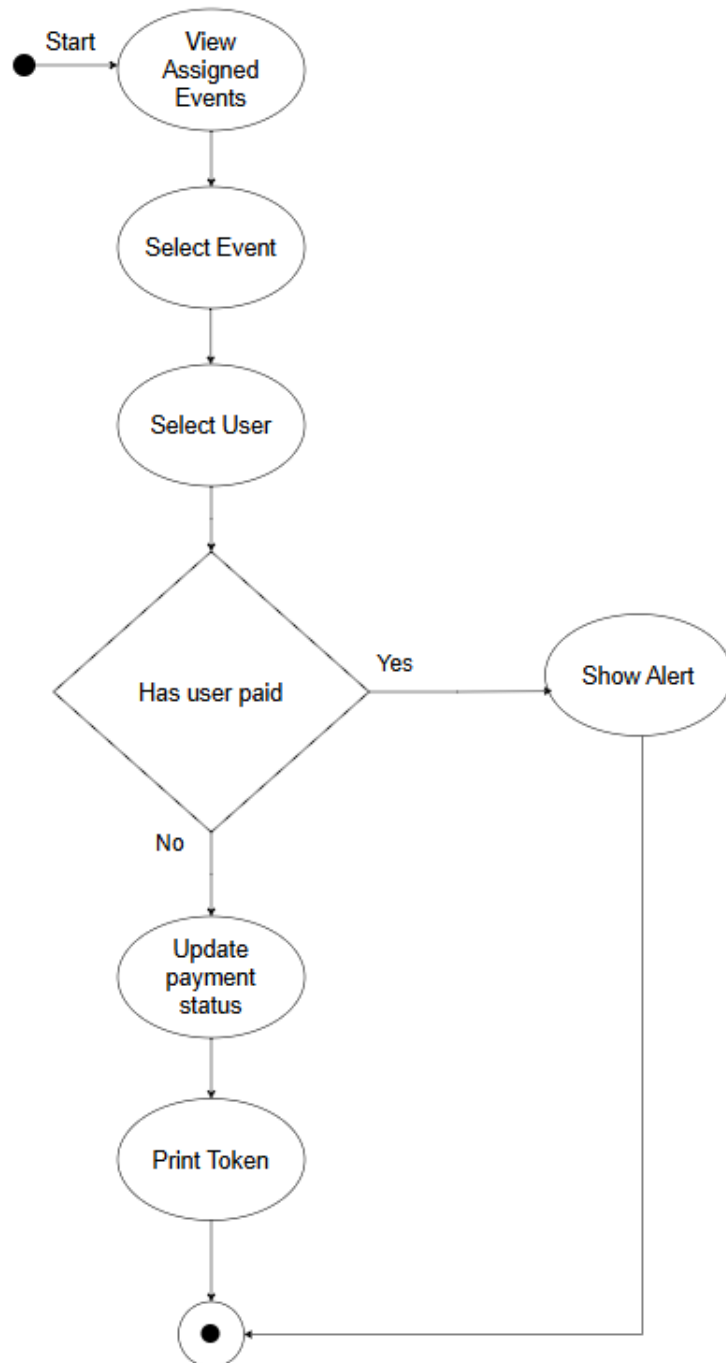


Figure 4.6: Activity Diagram for Toggle Event Participant Payment Status

4.2.7 Activity Diagram for Participant Token Scan

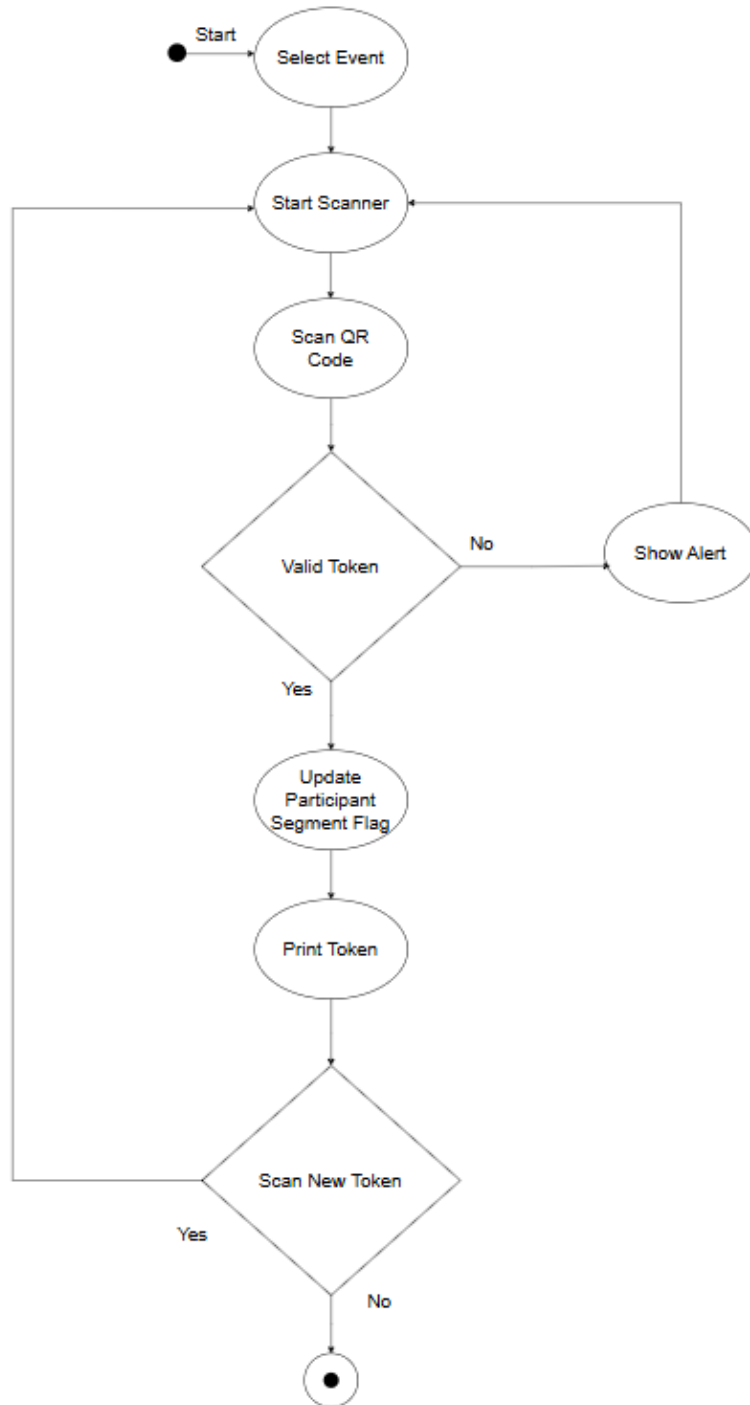


Figure 4.7: Activity Diagram for Participant Token Scan

4.2.8 Activity Diagram for Event Creation

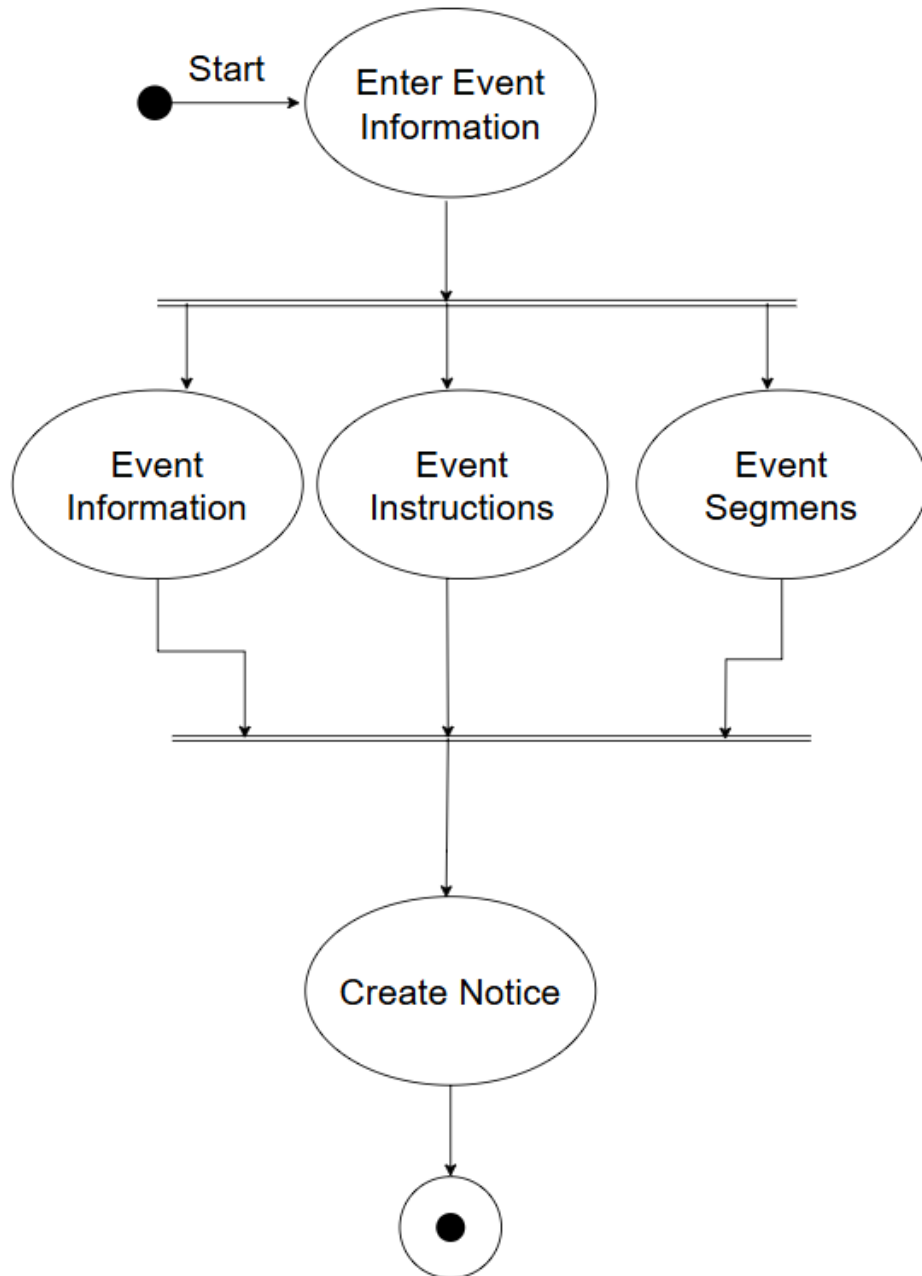


Figure 4.8: Activity Diagram for Participant Token Scan

4.2.9 Activity Diagram for Executive Creation

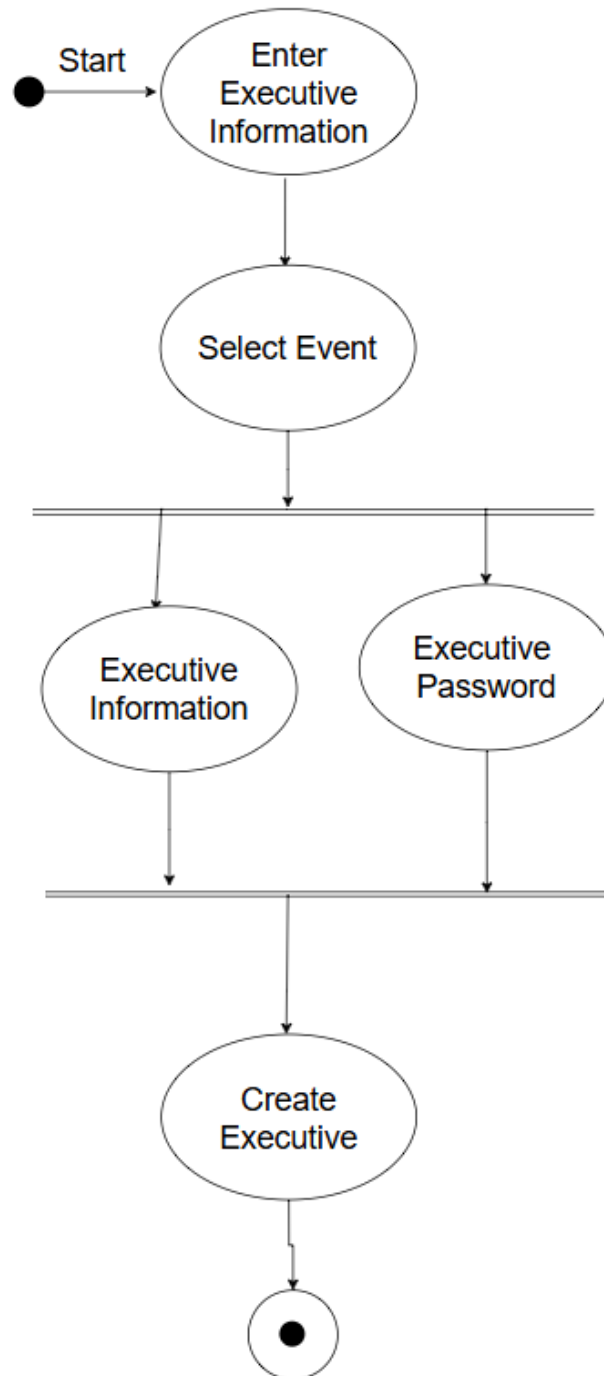


Figure 4.9: Activity Diagram for Executive Creation

4.2.10: Activity Diagram for Coordinator Creation

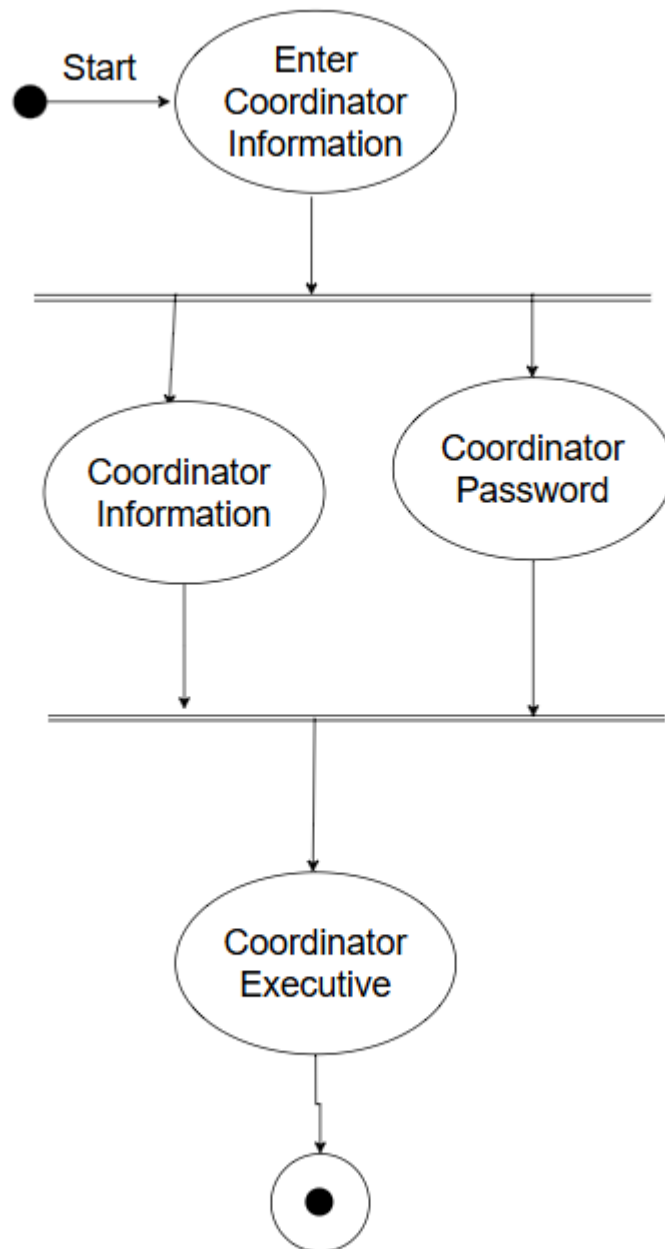


Figure 4.10: Activity Diagram for Coordinator Creation

4.2.11: Activity Diagram for AO Creation

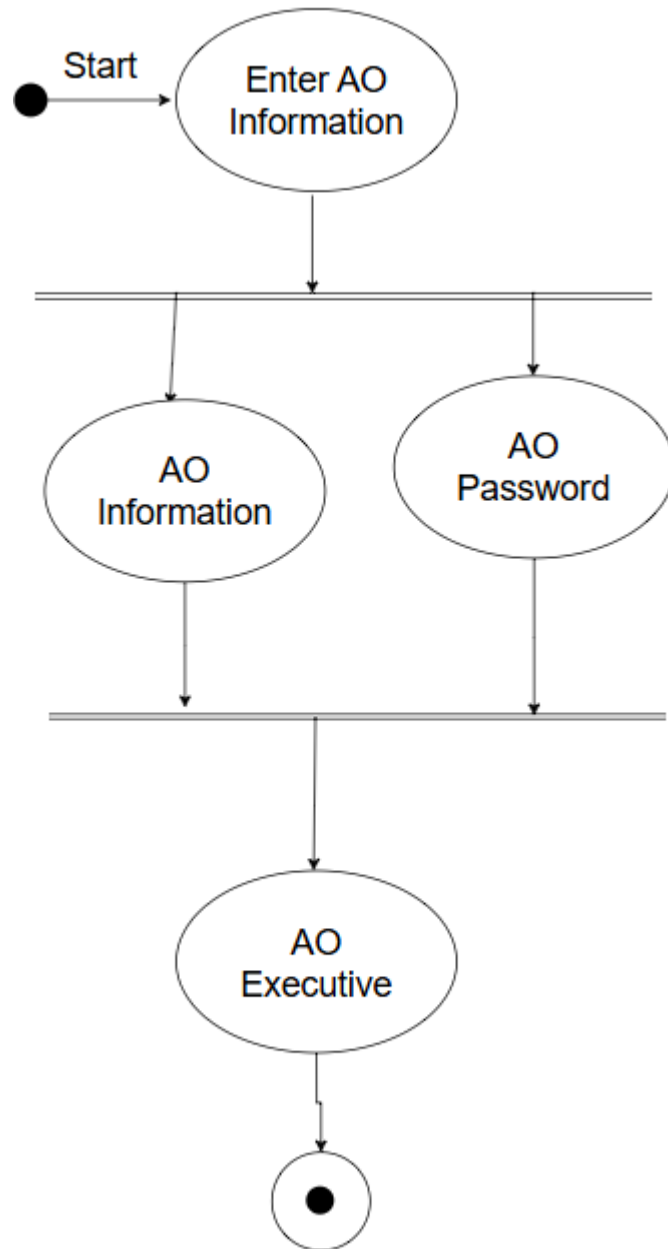


Figure 4.11: Activity Diagram for AO Creation

4.3 Swimlane Diagram

A swimlane diagram is a type of UML activity diagram that organizes activities into separate lanes, each representing a different actor, role, or organizational unit involved in a process. These lanes help visually distinguish who is responsible for each step in a workflow, making it easier to understand the flow of control and responsibilities across different parts of a system.

Swimlane diagrams resemble flowcharts but are structured horizontally or vertically to show interactions and dependencies between multiple participants. By combining sequential logic with role-based partitioning, swimlane diagrams offer a clear, organized view of complex processes, making them ideal for analyzing workflows, improving coordination, and identifying process inefficiencies.

4.3.1 Swimlane Diagram for Teacher Counselling Request

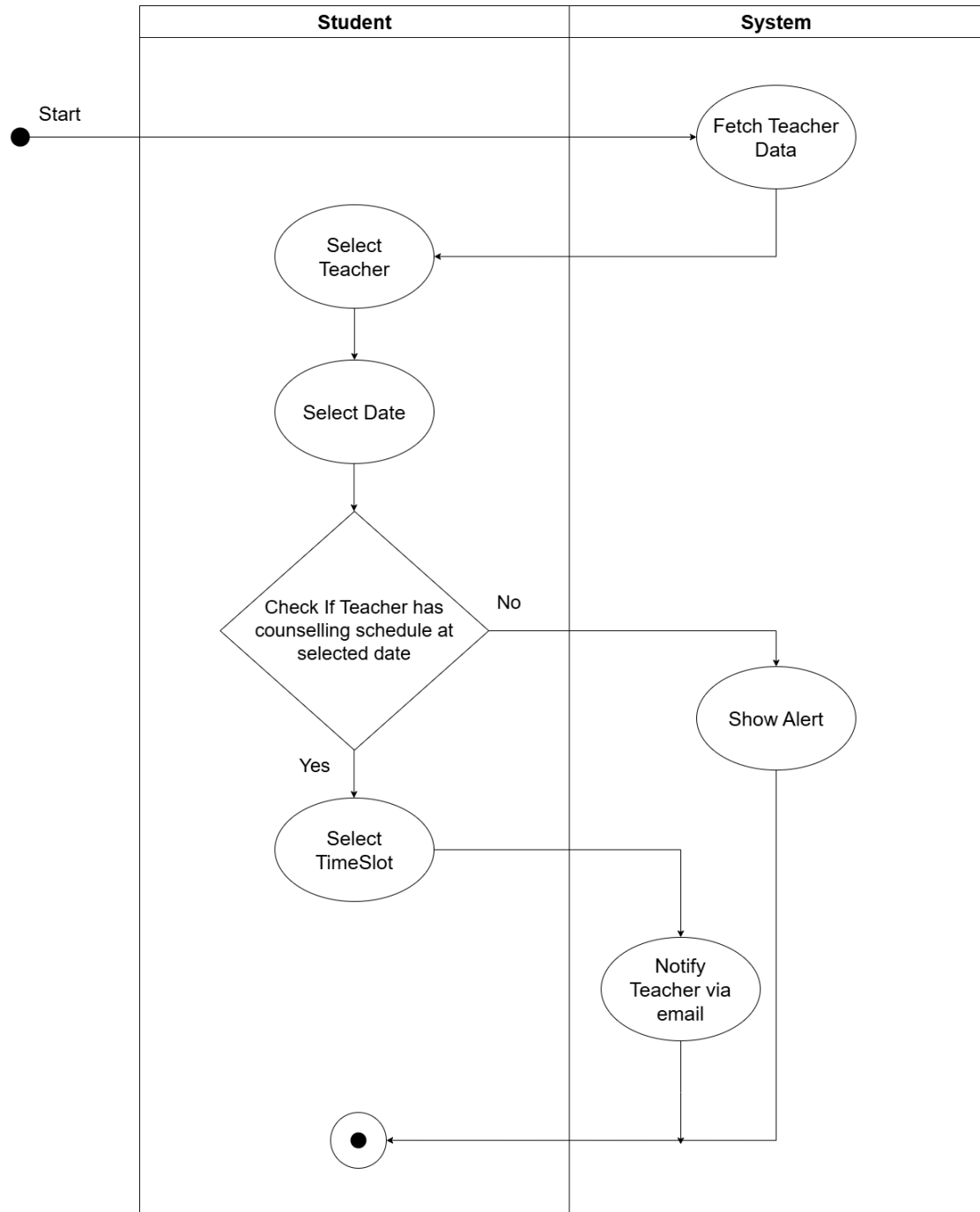


Figure 4.12: Swimlane Diagram for Teacher Counselling Request

4.3.2 Swimlane Diagram for Student Practicum Request

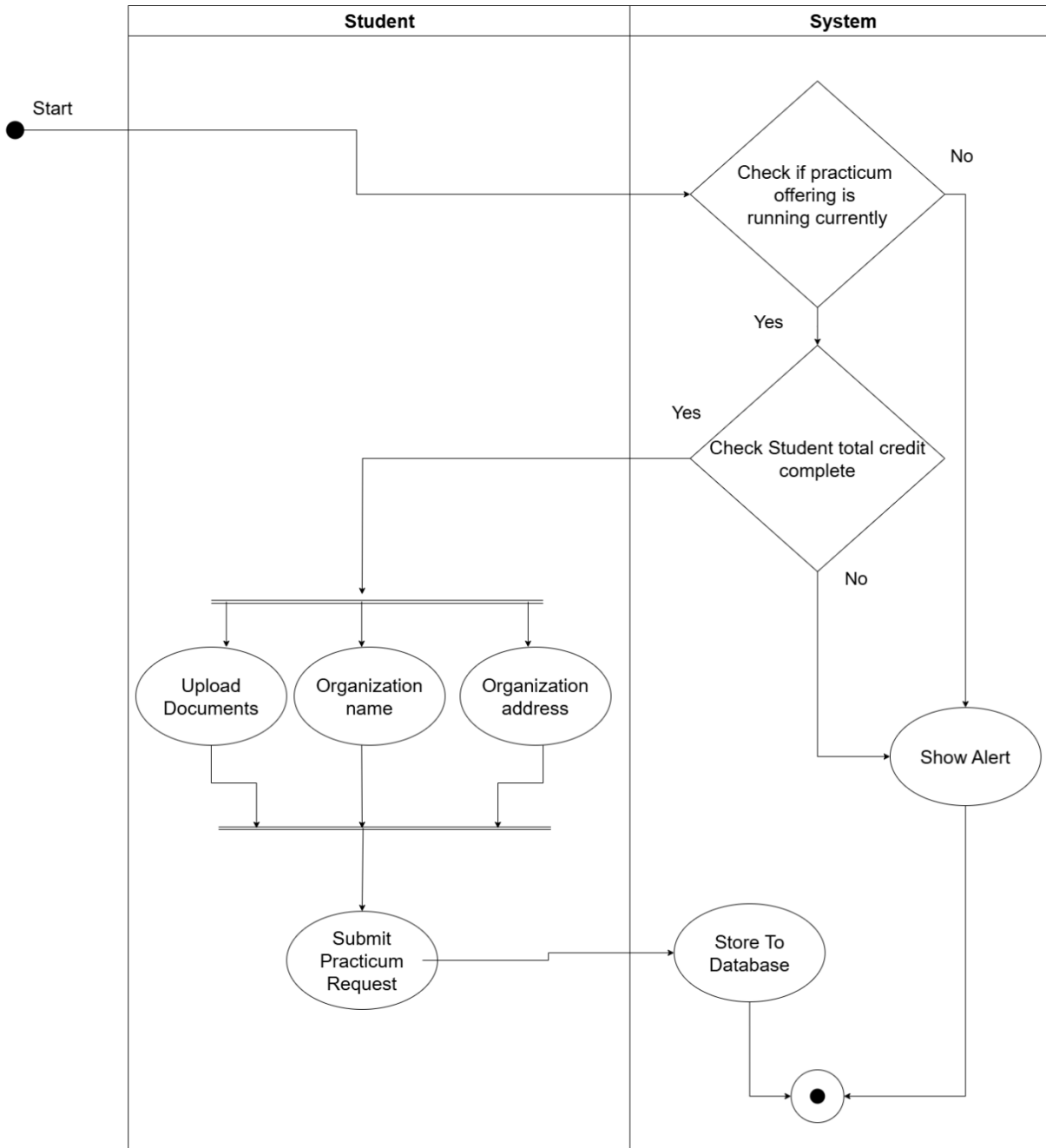


Figure 4.13: Swimlane Diagram for Student Practicum Request

4.3.3 Swimlane Diagram for Event Registration

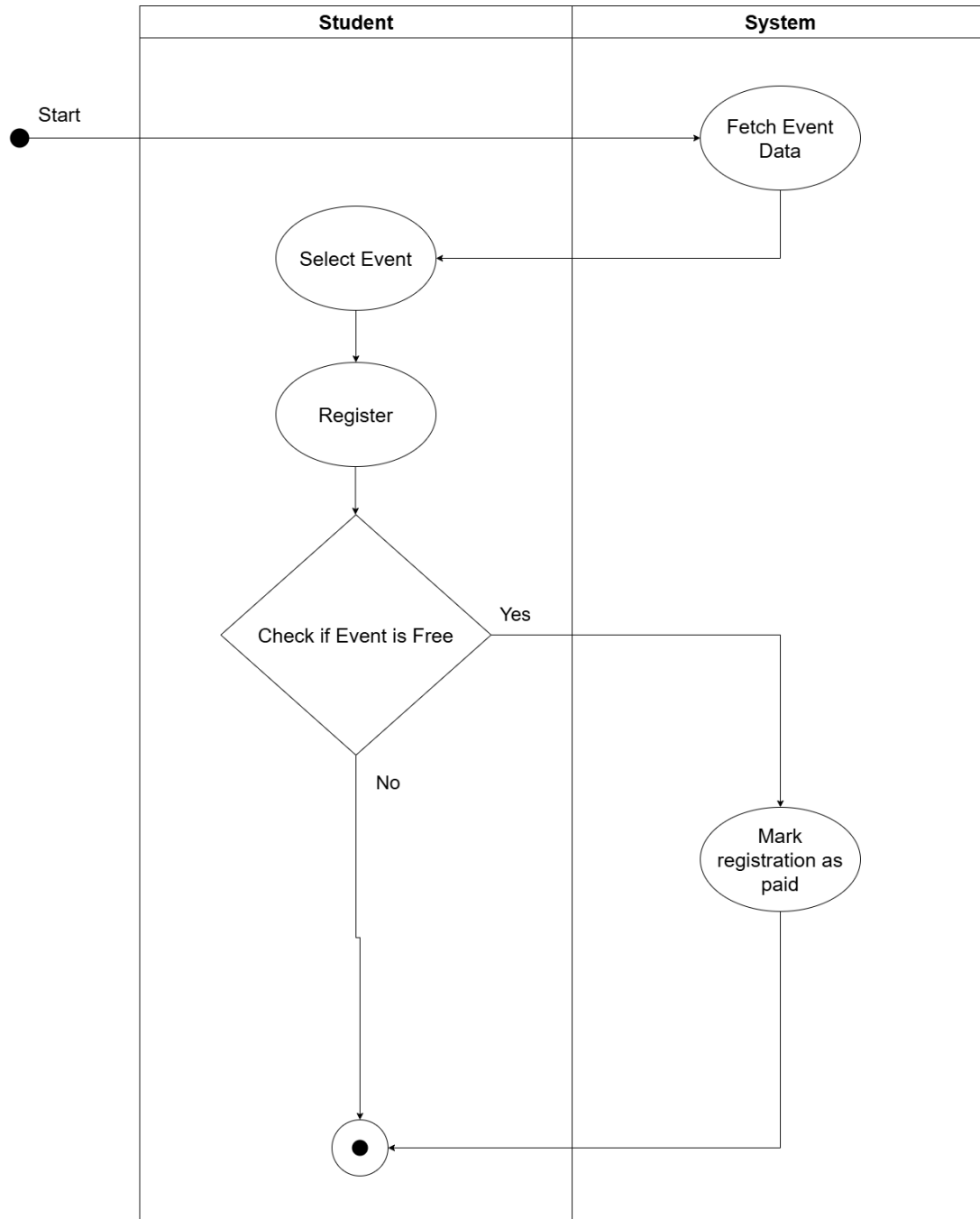


Figure 4.14: Swimlane Diagram for Event Registration

4.3.4 Swimlane Diagram for Creating Teacher Counselling Schedule

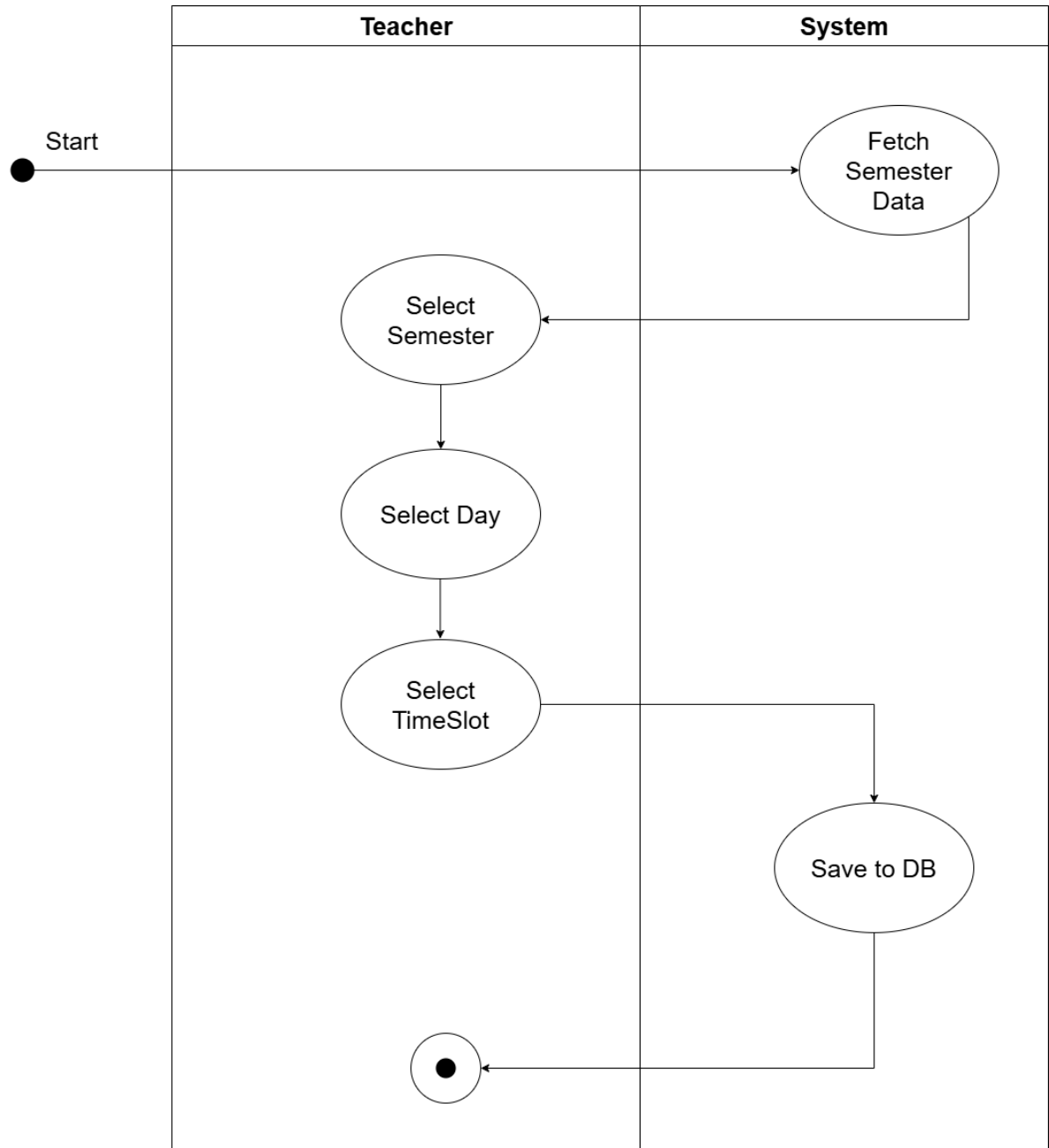


Figure 4.15: Swimlane Diagram for Creating Teacher Counselling Schedule

4.3.5 Swimlane Diagram for Executive Mobile App Login

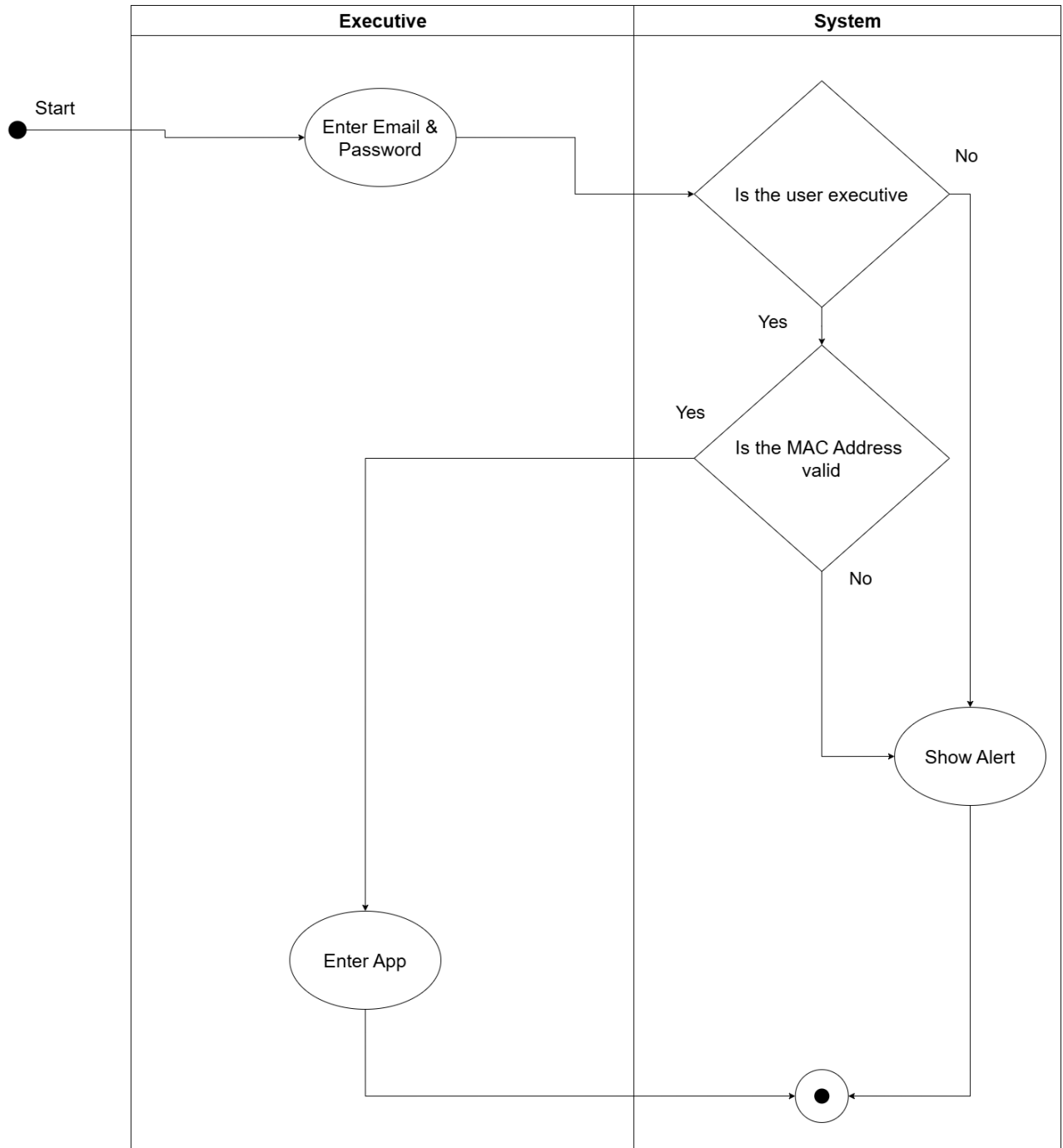


Figure 4.16: Swimlane Diagram for Executive Mobile App Login

4.3.6 Swimlane Diagram for Toggle Event Participant Payment Status

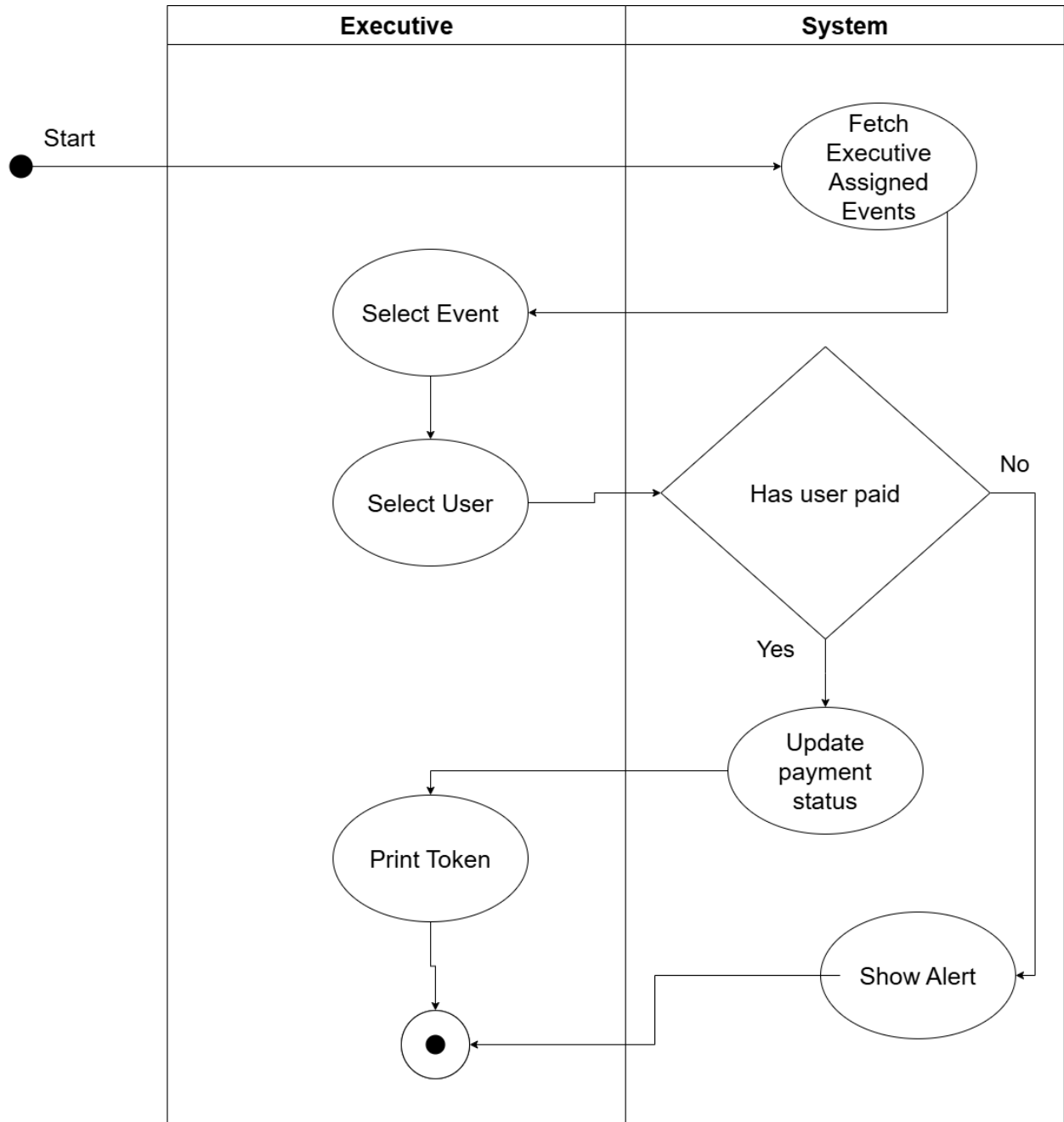


Figure 4.17: Swimlane Diagram for Toggle Event Participant Payment Status

4.4 Class Diagram

A class diagram is a type of UML (Unified Modeling Language) diagram used in software engineering to visually represent the structure of a system by showing its classes, attributes, methods, and the relationships among objects.

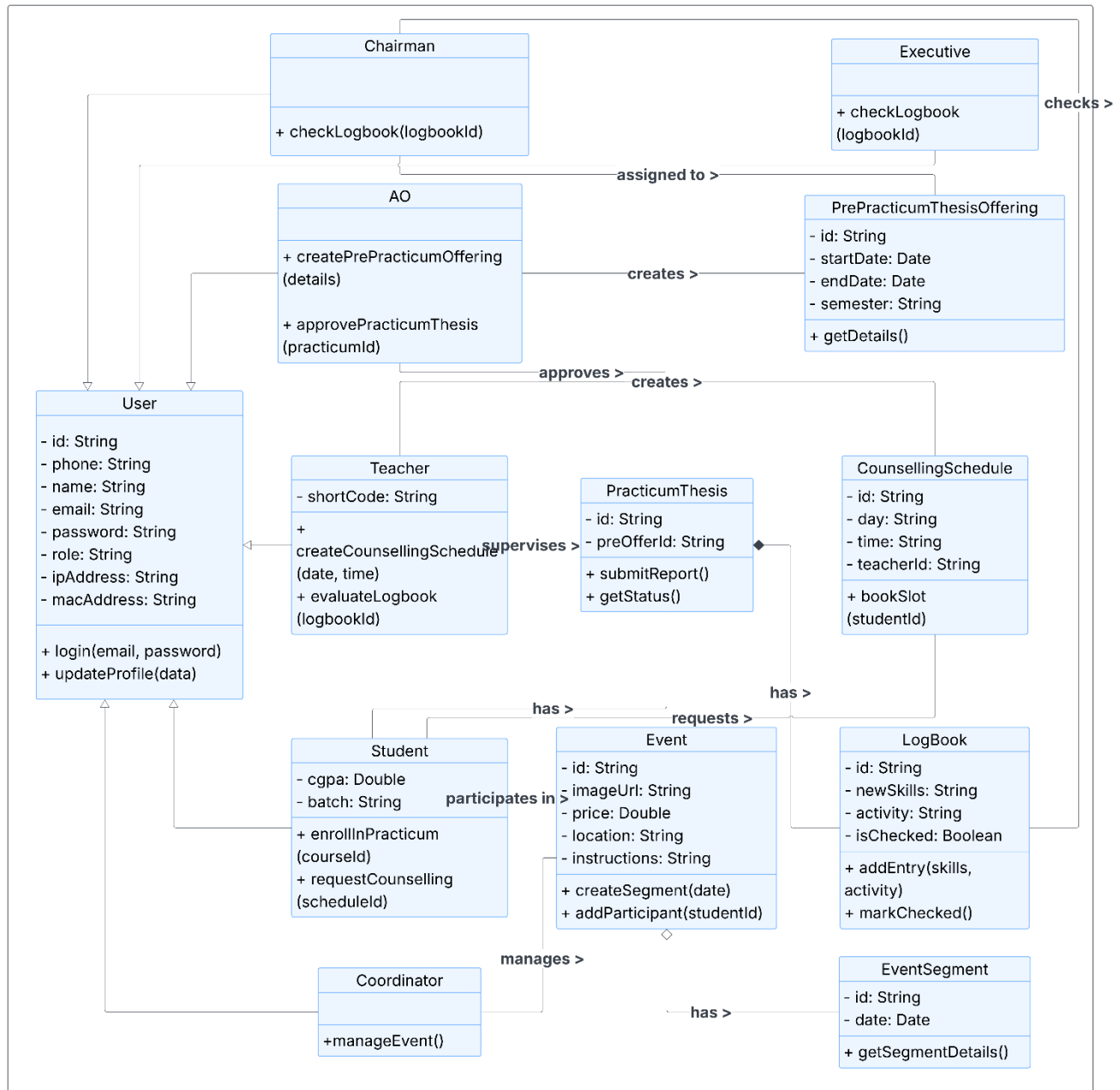


Figure 4.18: Class Diagram

Chapter 5

System Planning

5.1 Functions of Proposed System

The following table outlines the primary functions of the proposed Automated Administrative Services Management System at IUBAT.

Table 5.1 Functions of Proposed System

Function	Code
Registration	F01
Login	F02
Manage Users	F03
Manage Notice	F04
Manage Pre-Practicum Offer	F05
Manage Pre-Thesis Offer	F06
Manage Log Book	F07
View Notices	F08
Room Allocation	F09
Generate Sections	F10
Add Course Information	F11
Add Teacher Information	F12
View Routine	F13

5.1.1 Function Description

Function descriptions provide a comprehensive overview of each function including inputs, outputs, and the database tables involved.

- **Create New User (Registration)**
 - Input: User information (name, email, password)
 - Output: Successful registration message or error message
 - Database table: users

- **Login**
 - Input: Email and password
 - Output: Access to the system if credentials are valid, otherwise validation error message
 - Database table: users
- **Manage Users**
 - Input: User information (name, role, status)
 - Output: Successful addition, update, or error message
 - Database table: users
- **Manage Courses**
 - Input: Course information (course name, code, description, credits)
 - Output: Successful addition, update, or error message
 - Database table: courses
- **Manage Semesters**
 - Input: Semester information (semester name, start date, end date)
 - Output: Successful addition, update, or error message
 - Database table: semesters
- **Create Timetables**
 - Input: Timetable details (course, date, time, instructor)
 - Output: Successful timetable creation or error message
 - Database table: timetables
- **Handle Requests**
 - Input: Request details (type, user ID, description)

- Output: Successful request handling or error message
 - Database table: requests
- **Provide Feedback**
 - Input: Feedback details (course, instructor, feedback)
 - Output: Successful feedback submission or error message
 - Database table: feedback
- **View Routine**
 - Input: User ID
 - Output: Routine details or "not found" message
 - Database table: routines
- **Enroll in Courses**
 - Input: Enrollment request details (user ID, course ID)
 - Output: Successful enrollment or error message
 - Database table: enrollments
- **Apply for Thesis/Practicum**
 - Input: Application details (user ID, thesis/practicum details)
 - Output: Successful application or error message
 - Database table: applications
- **Participate in Pre-Course Offering**
 - Input: Suggested courses, student preferences
 - Output: Successful participation or error message
 - Database table: pre-course offerings

- **Submit Requests**

- Input: Request details (user ID, type, description)
- Output: Successful request submission or error message
- Database table: requests

5.2 System Project Planning

The software project planning for the Automated Administrative Services Management System includes several key activities:

- System Project Estimation
- Function Oriented Metrics
- Process Based Estimation
- Effort Distribution
- Task Scheduling
- Project Schedule Chart
- Cost Estimation

5.2.1 System Project Estimation

The accuracy of a software project estimate depends on several factors:

- Accurate estimation of the product size.
- Skill in converting the estimated size into monetary value, time, and man-hours.
- Accurate representation of the team's technical skills in the project plan.
- Consistency of product specifications and the environment supporting the software engineering effort.

5.2.2 Function Oriented Metrics

To compute function points (FP), the following steps and formulas are used:

1. Identify Complexity for Transaction Functions

Table 5.2 Identifying Complexity for Transaction Functions

Transaction Functions	Fields/ File Involvement	FTRs	DETs
User Registration (EI)	Fields: Name, Email, Password, Confirm Password File: User Database	1	4
Login (EI)	Fields: Email, Password File: User Database	1	2
Course Management (EQ)	Fields: Course ID, Course Name, Instructor File: Course Database	1	3
Request Handling (EQ)	Fields: Request ID, Request Type, Request Status File: Request Database	1	3
Admin User Management (EQ)	Fields: User ID, User Name, User Role File: User Database	1	3
Semester Management (EQ)	Fields: Semester ID, Semester Name, Start Date, End Date File: Semester Database	1	4
Report Generation (EO)	Fields: Report ID, Report Type, Generation Date File: Report Database	1	3

2. Identify Complexity for Data Functions

Table 5.3 Identifying Complexity for Data Functions

Data Functions	Fields/ File Involvement	RETs	DETs
User Database (ILF)	User ID, Name, Email, Password, Role	1	5
Course Database (ILF)	Course ID, Course Name, Instructor	1	3
Request Database (ILF)	Request ID, Request Type, Request Status	1	3
Semester Database (ILF)	Semester ID, Semester Name, Start Date, End Date	1	4
Report Database (ILF)	Report ID, Report Type, Generation Date	1	3

3. Calculate Unadjusted Function Points (UFP)

3.1 Unadjusted Function Point Calculation for Transaction Functions

Table 5.4 Unadjusted Function Point Contributions

Transaction Functions	FTRs	DETs	Complexity	UFP
User Registration (EI)	1	4	Low	3
Login (EI)	1	2	Low	3
Course Management (EQ)	1	3	Low	3
Request Handling (EQ)	1	3	Low	3
Admin User Management (EQ)	1	3	Low	3
Semester Management (EQ)	1	4	Low	3
Report Generation (EO)	1	3	Low	4
Total				22

3.2 Unadjusted Function Point Calculation for Data Functions

Table 5.5 Unadjusted Function Point Calculation for Data Functions

Data Functions	RETs	DETs	Complexity	UFP
User Database (ILF)	1	5	Low	7
Course Database (ILF)	1	3	Low	7
Request Database (ILF)	1	3	Low	7
Semester Database (ILF)	1	4	Low	7
Report Database (ILF)	1	3	Low	7
Total				35

4. Performance and Environmental Impact

Table 5.6 Performance and Environmental Impact

GSC	DI
Data Communication	2
Distributed Data Processing	1
Performance	3
Heavily Used Configuration	2
Transaction Rate	2
Online Data Entry	3
End-User Efficiency	5
Online Update	5
Complex Processing	1
Reusability	3
Installation Ease	5
Operational Ease	5
Multiple Sites	0
Facilitate Change	4
Total Degree of Influence	41

3. Counting Adjusted Function Points

- $UFP = UFP(\text{Transaction Fn}) + UFP(\text{Data Fn}) = 22 + 35 = 57$
- $\text{Value Adjustment Factor (VAF)} = 0.65 + (0.01 * TDI) = 0.65 + (0.01 * 41) = 1.06$
- $\text{Adjusted Function Points (AFP)} = UFP * VAF = 57 * 1.06 = 60.42$

5.2.3 Effort Distribution

- Hours Per Function Point for Dot Net : 15.5
- $\text{Effort for Dot Net} = AFP * \text{Productivity} = 60.42 * 15.5 = 936.51 \text{ hours}$
- $\text{Person Days} = 936.51 / 8 \text{ hours per day} \approx 117 \text{ days}$

Estimated work units for software development can be divided using a realistic effort distribution model:

Effort Allocation:

- Requirement Analysis & Planning: 45%
- System Design: 25%
- System Development: 20%
- Testing: 5%
- Deployment: 5%

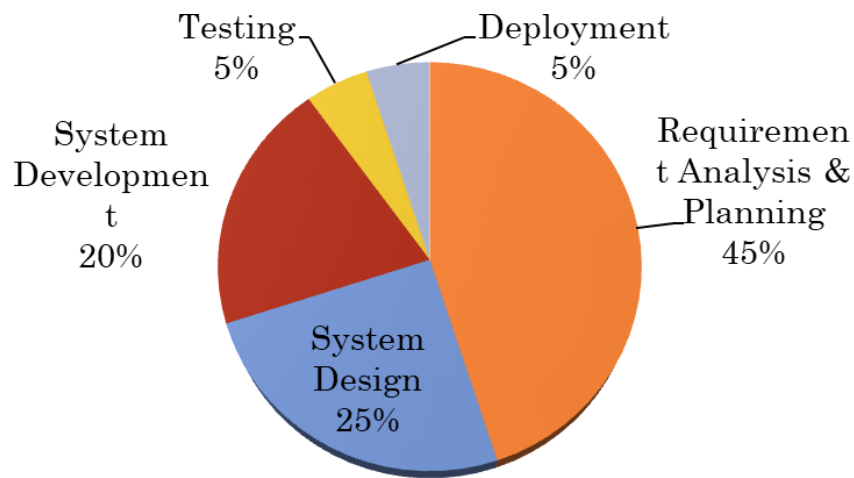


Figure 5.1 Effort Distribution

5.2.4 Task Scheduling

Project scheduling distributes estimated efforts across the planned duration of the project, adhering to principles such as:

- Compartmentalization: Breaking the project into manageable tasks.
- Interdependency: Assessing the interdependence of each task.
- Time Allocation: Allocating specific work units to each task.

5.2.5 Project Schedule Chart

A strict timeline ensures timely and sequential completion of interdependent tasks.

Table 5.7 Project Schedule Chart

Task	Duration (Weeks)
Requirement Analysis	8
System Design	5
System Development	3
System Testing	2
Total Duration	18

Tasks	1st Week	2nd Week	3rd Week	4th Week	5th Week	6th Week	7th Week	8th Week	9th Week	10th Week	11th Week	12th Week	13th Week	14th Week	15th Week	16th Week	17th Week	18th Week
Requirement Analysis																		
System Designing																		
System Development																		
System Testing																		

Figure 5.2 Project Schedule Chart

5.2.6 Cost Estimation

Software cost estimation is based on personnel, hardware, software, and other costs.

5.2.6.1 Personnel Cost

Table 5.8 Personnel Cost Estimation

Designation	Number of People	Working Hours	First Payment (60%)	Remaining Payment (40%)	Weekly Payment	Total Salary
System Designer	1	936.51	18,432	12,288	768	30,720
Coder	2	936.51	24,576	16,384	1,024	40,960
Total						71,680

5.2.6.2 Hardware Cost

Table 5.9 Hardware Cost

Hardware	Quantity	Unit Cost (TK)	Total Cost (TK)
PC	1	4,270	4,270
Scanner	1	160	160
Printer	1	180	180
Total Hardware Cost			4,610

5.2.6.3 Software Cost

Table 5.10 Software Cost

Software	Cost (TK)
Microsoft Office 2022	3,500
Windows 11	10,000
Visual Studio Code	Free
SQL Server	Free
Total Software Cost	13,500

5.2.6.4 Other Cost

Table 5.11 Other Cost

Item	Quantity	Unit Cost (TK)	Total Cost (TK)
Pen and Paper	-	700	700
Internet Connection	-	100	100
Mobile	1	2,000	2,000
Total Other Costs			2,800

5.2.6.5 Accounts Table

Table 5.12 Accounts Table

Particulars	Cost (TK)
Salary:	
- System Designer	30,720
- Coder	40,960
Total Salary	71,680
Hardware Cost:	
- PC	4,270
- Scanner	160
- Printer	180
Total Hardware Cost	4,610
Software Cost:	
- Microsoft Office 2022	3,500
- Windows 11	10,000
- Visual Studio Code	Free
- SQL Server	Free
Total Software Cost	13,500
Other Costs:	
- Pen and Paper	700
- Internet Connection	100
- Mobile	2,000
Total Other Costs	2,800
Total Project Cost	92,590

By accurately planning and estimating resources, time, and costs, the Departmental Course Management System project aims to achieve efficiency and effectiveness in managing courses, users, and related activities at IUBAT

Chapter 6

Risk Engineering

6.1 Risk Management Overview

A risk is an uncertain circumstance that could lead to an issue. It is crucial to assess the risks associated with a project to mitigate potential problems. Failure to carefully evaluate and estimate risks in a software project can result in various challenges. By employing risk analysis and management techniques, a software development team can better understand and address uncertainties.

6.1.1 Risk Identification

Table 6.1 Risk Identification

Risk Type	Possible Risk
Technology	- The server has limited capacity. If exceeded, the server might crash.
	- Future editing is not maintainable.
	- Database has no backup.
People	- Team members are unavailable or ill at critical times.
	- Members lack the knowledge to use the needed tools.
Organizational	- The amount of financial support needed might not be met.
Tools	- The code cannot be generated fully.
	- The tools used might be outdated.
Requirements	- Major changes to the established requirements might be added.
Estimation	- Time to create the project is underestimated.
	- The rate of defect repair is underestimated.
	- The size of the software is underestimated.

6.1.2 Risk Analysis

Table 6.2 Risk Analysis

Risk	Probability	Effects
The amount of financial support needed might not be met.	Low	Catastrophic
Database has no backup.	Low	Catastrophic
The server has limited capacity.	Moderate	Serious

Team members are unavailable or ill at critical times.	Moderate	Serious
Major changes to the established requirements might be added.	Moderate	Serious
The rate of defect repair is underestimated.	Moderate	Tolerable
Time to create the project is underestimated.	Low	Tolerable
The size of the software is underestimated.	Moderate	Tolerable
The tools which are used might be outdated.	Low	Tolerable
Members lack the knowledge to use needed the tools.	High	Tolerable
The code cannot be generated fully.	High	Insignificant
Future editing is not maintainable.	High	Insignificant

6.1.3 Risk Planning

Table 6.3 Risk Planning

Risk	Strategy
The amount of financial support needed might not be met.	A backup fund is prepared, and additional funding sources are explored.
Database has no backup.	Investigate for a backup database.
The server has limited capacity.	Investigate for a better-performing server.
Team members are unavailable or ill at critical times.	Make other team members familiar with critical tasks to ensure continuity.
Major changes to the established requirements might be added.	Be flexible and prepared to adapt to changes in requirements.
The time to create the project is underestimated.	Create a backup schedule to account for potential delays.
The size of the software is underestimated.	Explore options for additional storage capacity.
The tools used might be outdated.	Investigate if updated versions or alternative tools are available.
Members lack the knowledge to use needed the tools.	Provide training or a crash course to improve skills.
Future editing is not maintainable.	Develop detailed documentation to facilitate future edits and maintenance.

6.1.4 The RMMM Plan

6.1.4.1 Technology Risk

Table 6.4 Technology Risk

Name	Lack of security, service, and defective components
Probability	Moderate
Impact	Serious
Description	Inadequate security measures could result in data breaches, compromising sensitive data.
Mitigation & Monitoring	Use strong passwords or two-factor authentication, back up services, and check components before use.
Management	Utilize restore from backup in case of outages, and ensure the use of strong passwords.
Status	Managed

6.1.4.2 People Risk

Table 6.5 People Risk

Name	Lack of adequate skills or expertise
Probability	Moderate
Impact	Serious
Description	Some staff may not be trained to use required technology during project development.
Mitigation & Monitoring	Regular staff meetings to address turnover and working conditions.
Management	Acknowledge and monitor staff skills, intervene if necessary.
Status	Monitoring

6.1.4.3 Tool Risk

Table 6.6 Tools Risk

Name	Insufficient budget or required budget increased.
Probability	Moderate
Impact	Serious
Description	Insufficient budget may lead to incomplete project or incompatible software tools.
Mitigation & Monitoring	Explore alternative tools and conduct compatibility tests.
Management	Use alternative tools and adjust budget accordingly.
Status	Problem resolved

6.1.4.4 Estimation Risk

Table 6.7 Estimation Risk

Name	Insufficient time for development.
Probability	Moderate
Impact	Serious
Description	Underestimating development time leads to delays and missed deadlines.
Mitigation & Monitoring	Prepare a detailed schedule and allocate extra time for contingencies.
Management	Reschedule tasks to accommodate any delays and inform stakeholders in advance.
Status	Managed

Chapter 7

Designing

7.1 Structural Design

The goal of analysis modeling is to efficiently verify accuracy, completeness, and consistency by using a combination of textual and diagrammatic representations to communicate data, function, and behavior requirements in an easily understandable way. Here you will find UML resources and citations for both traditional and OOA approaches to analysis

7.2 Entity Relationship Diagram (ERD)

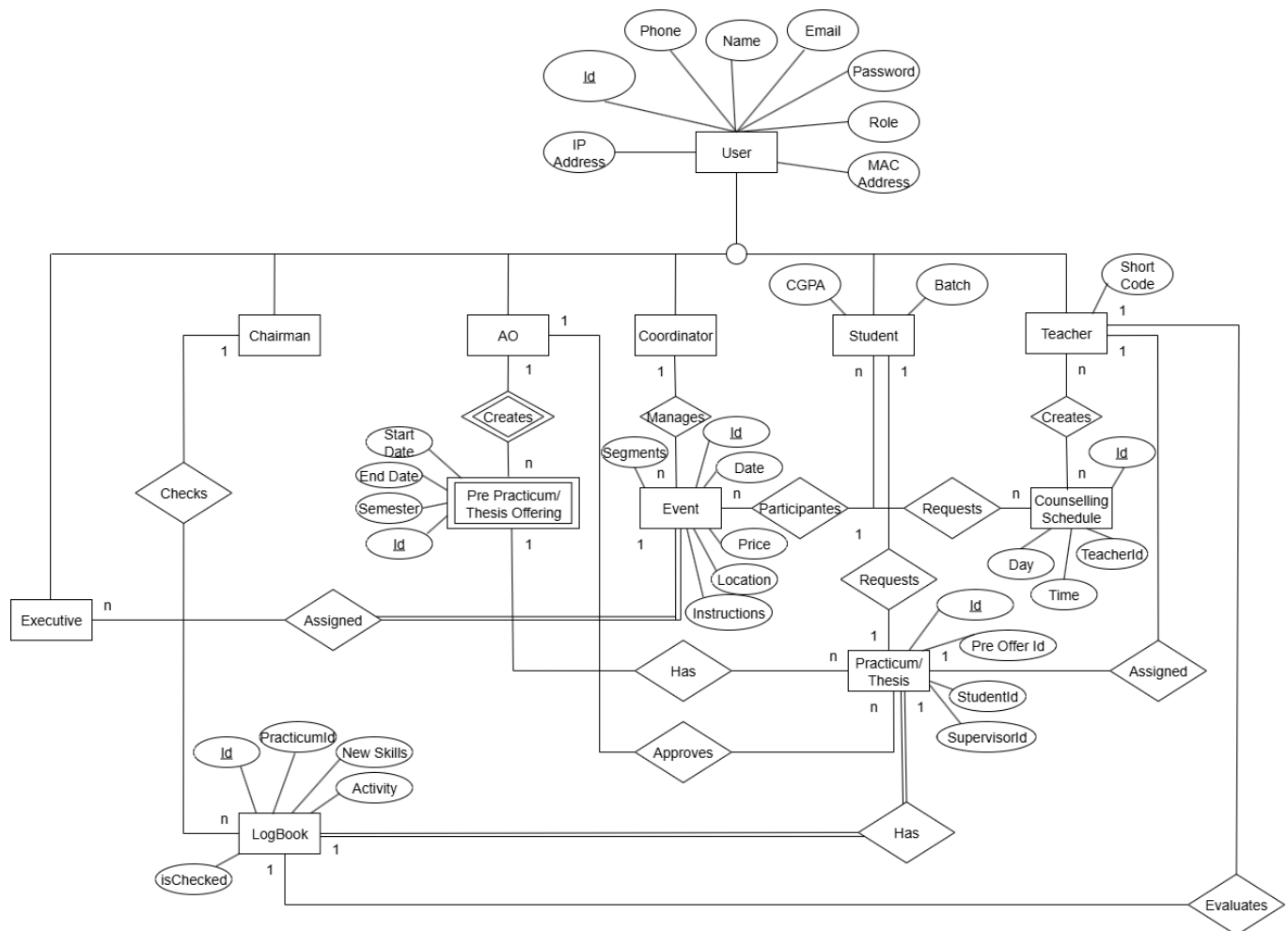


Figure 7.1: ER Diagram

A software engineer can use the Entity Relationship Diagram (ERD) to specify the data items that are input and output from a system, as well as the characteristics that

determine their qualities and relationships. It depicts the data structures and relationships in an amazing graphical format.

7.3 Architectural Design

In this project, a three-tier architecture is utilized. Each component of three-tier architecture—the user interface, presentation, data storage, and access—is designed and maintained independently. In addition to the standard advantages of modular software, the three-tier design allows for the independent upgrade or modification of any of the three layers in response to changes in needs. An RDBMS stores the logic for computer data storage, while a user interface runs on a desktop PC or other device. Functional process logic, which can be composed of one or more independent modules, operates on a workstation or application server. The following are the three levels that make up this design:

- **Presentation tier:** The top level of the application is the presentation layer. A web page or the graphical user interfaces of an operating system are examples of presentation tier elements that users may directly interact with. The results are sent to the browser/client tier and all other tiers of the network after it communicates with other levels.
- **Application tier:** The logical tier is separated from the display tier and functions as its own layer, controlling the functionality of an application by completing defined processing.
- **Database Layer:** Database servers, file sharing, and other data tier components are included. The data access layer should expose methods for handling stored data to the application tier via an Application Programming Interface (API), without revealing or creating dependence on the data storage mechanisms.

7.4 Data Flow Diagram

Data Flow Diagram (DFD) illustrates the kinds of information that will be entering and exiting the system, along with the sources and destinations of the data, as well as its storage locations. However, it does not provide details about process timing or whether processes will run sequentially or concurrently (which is typically depicted in a flowchart).

7.4.1 Context Level DFD

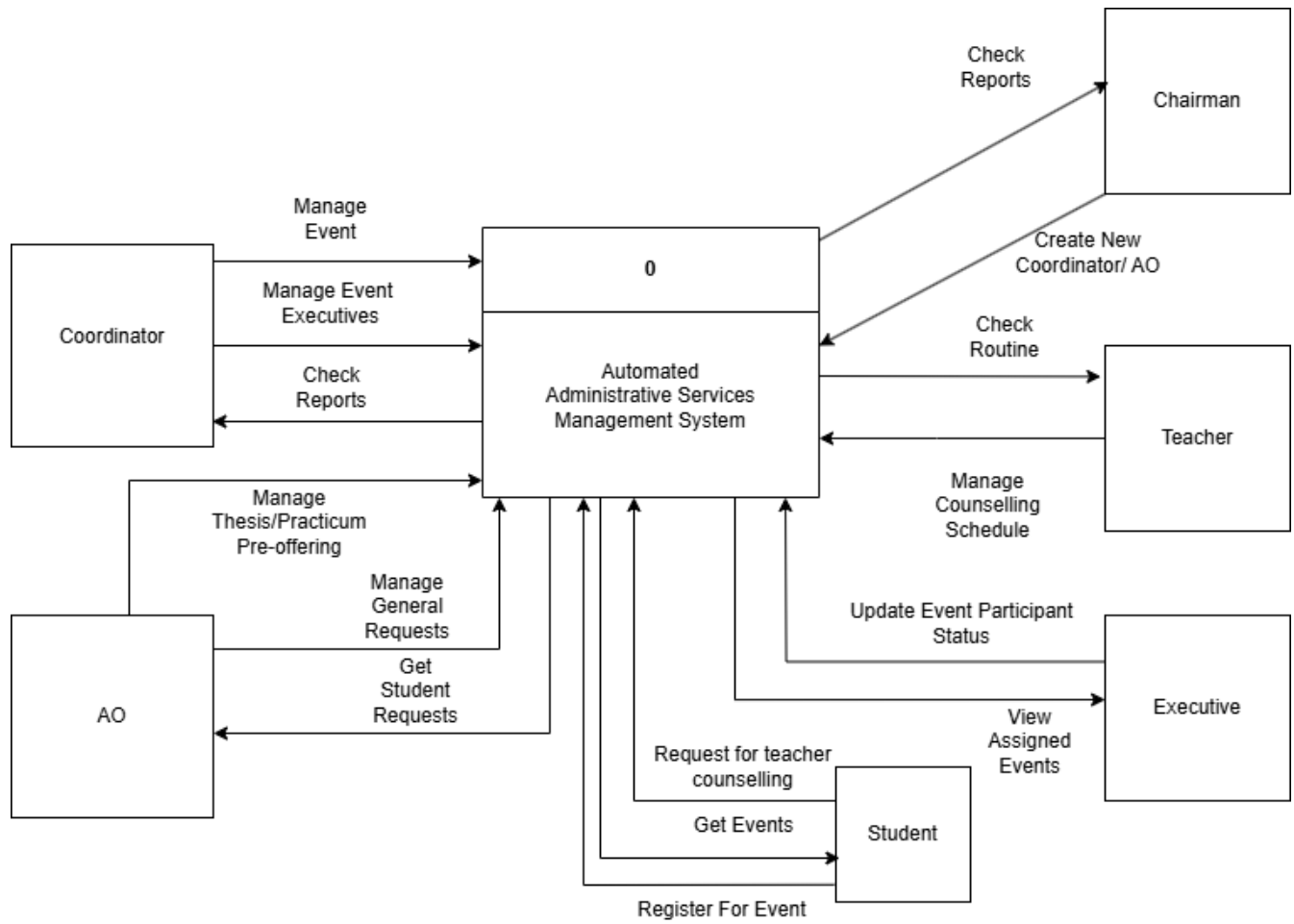


Figure 7.2: Level 0 DFD

7.4.2 Level 1 DFD

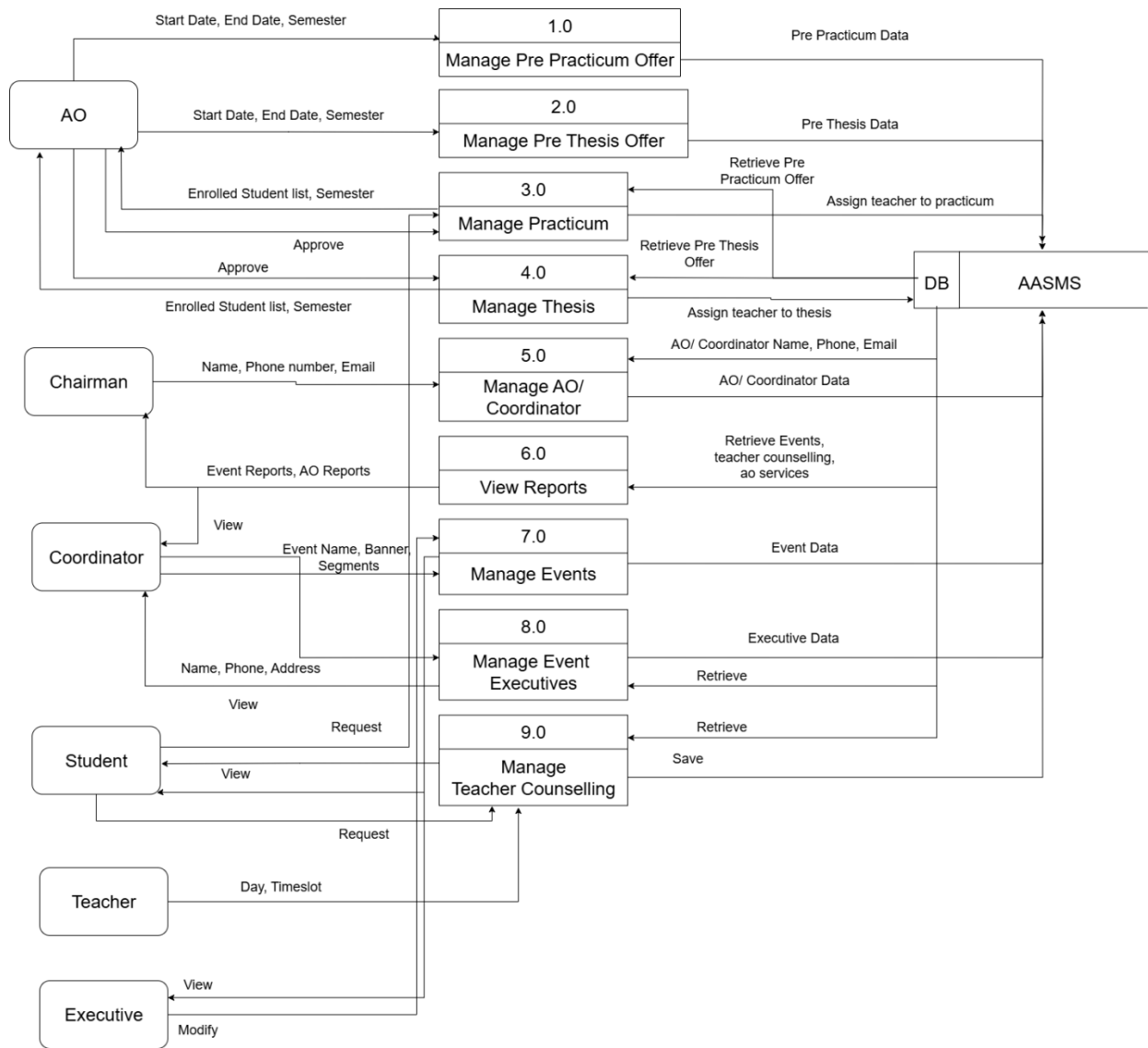


Figure 7.3: Level 1 DFD

7.4.3 Level 2 DFD

7.4.3.1 Level 2 DFD Process 1 (Pre-Thesis Offering)

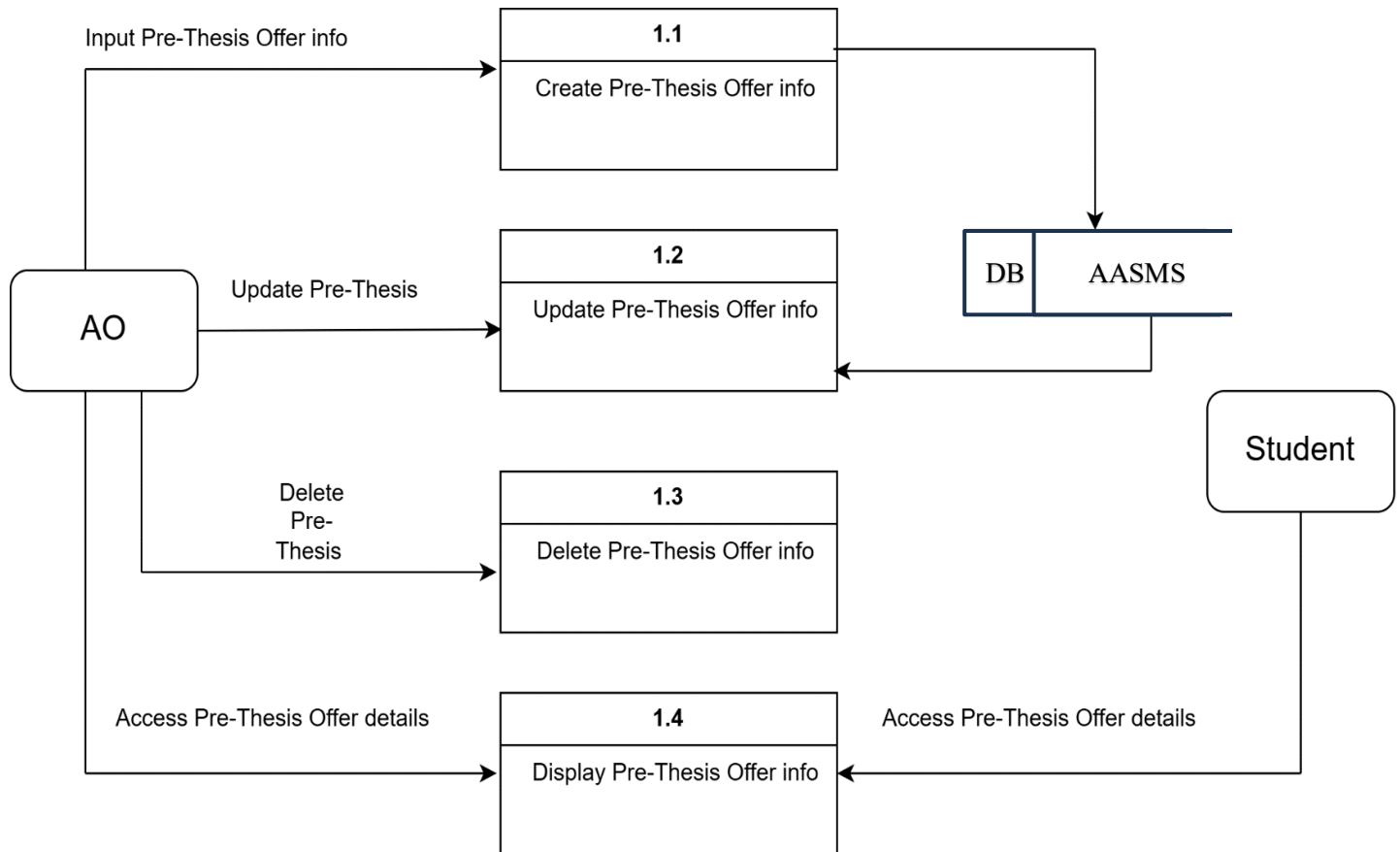


Figure 7.4: Level 2 Process 1 DFD

7.4.3.2 Level 2 DFD Process 3 (Notice)

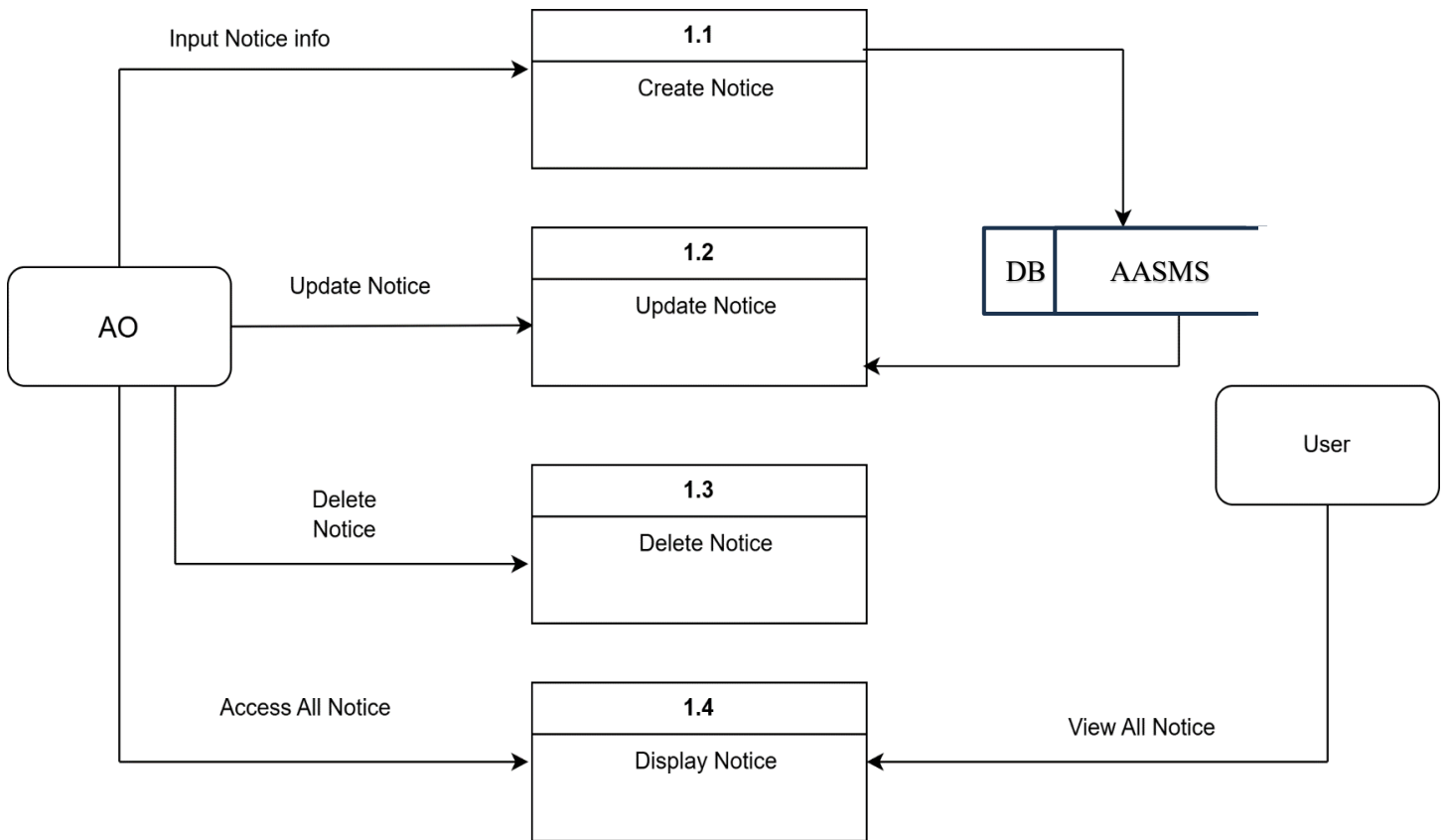


Figure 7.5: Level 2 Process 3 DFD

7.4.3.3 Level 2 DFD Process 4 (LogBook)

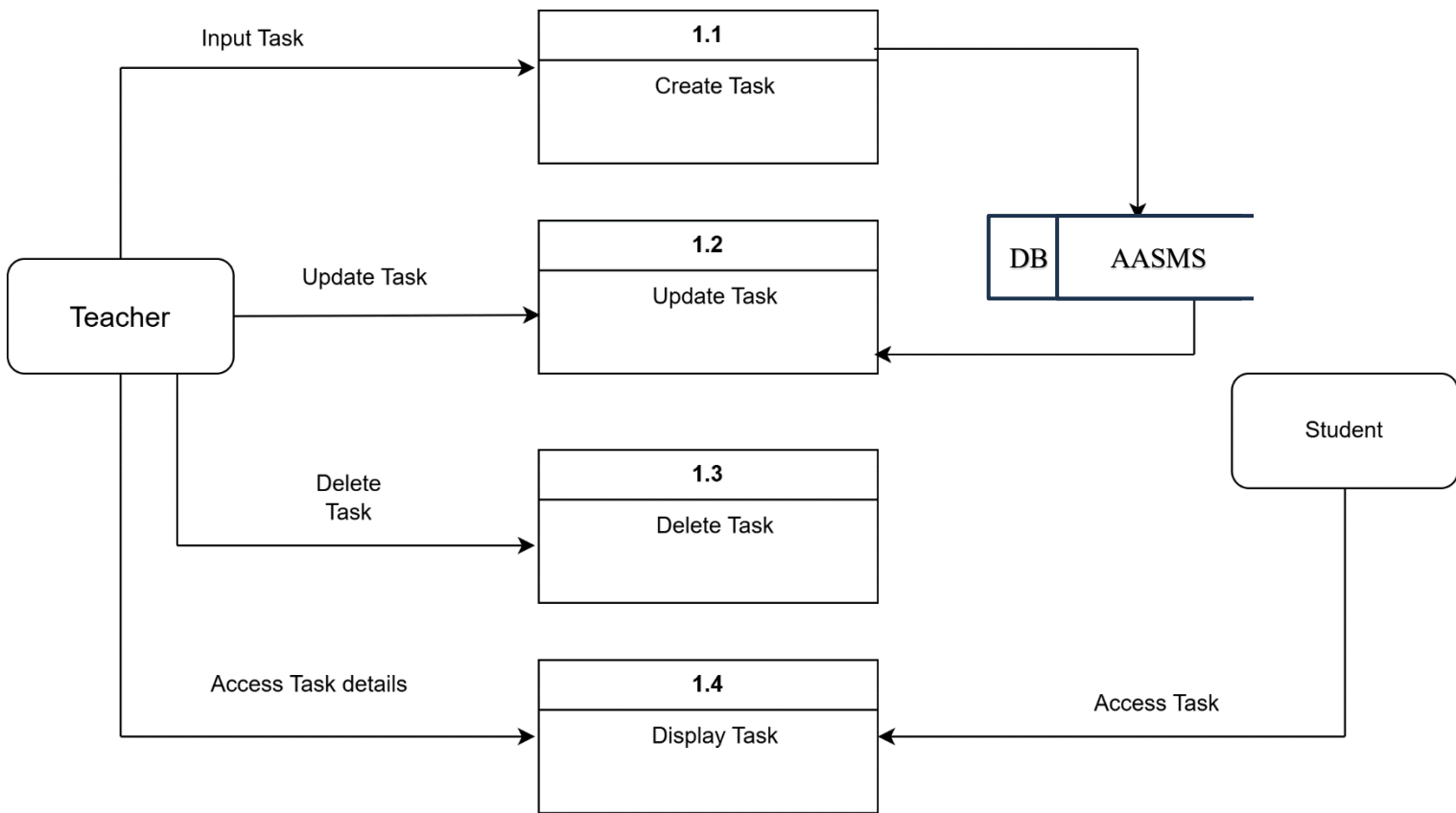


Figure 7.6: Level 2 Process 4 DFD

7.4.3.4 Level 2 DFD Process 5 (Routine)

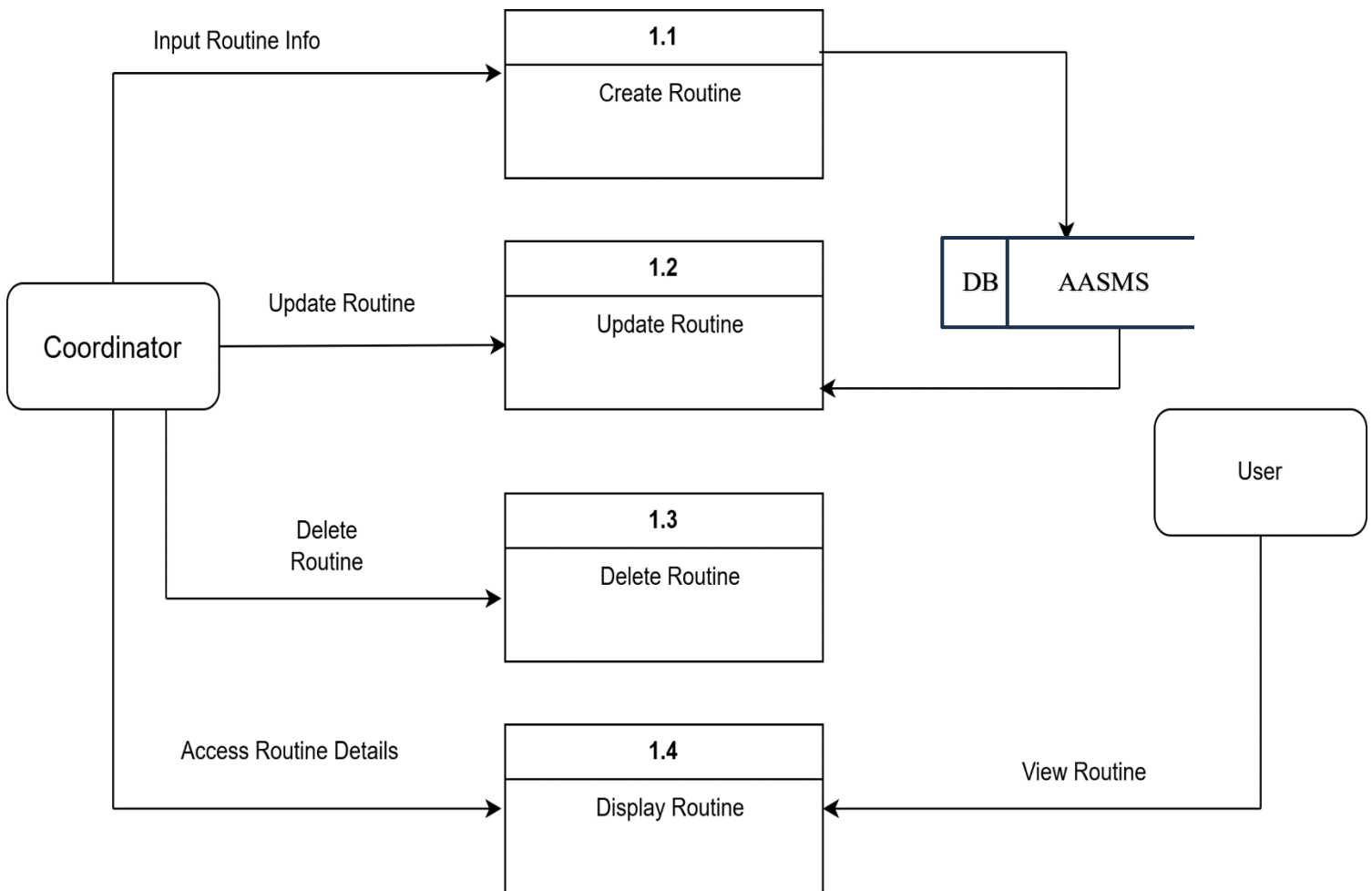


Figure 7.7: Level 2 Process 5 DFD

7.5 Interface of the system

7.5.1 Interface for Create Notice

**CSE Automation**

Logout 

Shariful Islam
shariful.ao@lubat.edu

Menus 

Students 

Teachers 

Semesters 

Batches 

Courses 

Time Slots 

Miscellaneous 

General Requests 

Add Notice Information

Notice Title *

Notice Description *

Notice Url

Select Notice User

All 

Add Notice

7.5.2 Interface for View All Notice

Shariful Islam

shariful.ao@iubat.edu

Menus

Students

Teachers

Semesters

Batches

Courses

Time Slots

Miscellaneous

General Requests

Pre Course Offering

Course Add Request

All Notice

Serial	Title	Description	Date	Action
1	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	
2	Practicum Offer	Practicum Offer for Fall 2024 Semester	Tue Mar 11 2025	
3	Practicum Offer	Practicum Offer for Summer 2024 Semester	Tue Mar 11 2025	
4	Meeting with Chairman sir	Regarding your practicum defense	Tue Mar 11 2025	
5	Meeting with chairman sir	For Practicum Board	Tue Mar 11 2025	
6	Ifter Mahfil CSE 2025	asdfsdf	Tue Mar 11 2025	
7	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	

7.5.3 Interface for Pre-Practicum Offering

 CSE Automation

Logout 

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Menus 

Students 

Teachers 

Semesters 

Batches 

Courses 

Practicum Offering

Select Semester *
Fall 2024 

Pre Offering Start *
04/23/2025 

Pre Offering End *
04/30/2025 

Submit

Shariful Islam
shariful.ao@iubat.edu

Menus 

Students 

All Notice

Serial	Title	Description	Date	Action
1	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	  

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7.5.4 Interface for Pre-Thesis Offering

 CSE Automation

Logout 

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Menu

Students

Teachers

Semesters

Batches

Courses

Time Slots

Thesis Offering

Select Semester *
Spring 2024

Pre Offering Start *
04/22/2025

Pre Offering End *
04/30/2025

Submit

 CSE Automation

Logout 

Shariful Islam
shariful.ao@iubat.edu

Menu

Students

All Notice

Serial	Title	Description	Date	Action
1	Thesis Offer	Thesis Offer for Spring 2024 Semester	Sat Apr 12 2025	  

7.5.5 Interface of a student for View All Notice

**CSE Automation**

Logout 

Shabit
shabitalahi123@gmail.com

General Requests 

Course Add Request 

Thesis Request

Practicum Meetings

All Practicum Logs

Add Practicum Log

Pre Course Offering

All Notice

Serial	Title	Description	Date	Action
1	Thesis Offer	Thesis Offer for Spring 2024 Semester	Sat Apr 12 2025	
2	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	
3	Practicum Offer	Practicum Offer for Fall 2024 Semester	Tue Mar 11 2025	
4	Practicum Offer	Practicum Offer for Summer 2024 Semester	Tue Mar 11 2025	
5	Meeting with Chairman sir	Regarding your practicum defense	Tue Mar 11 2025	
7	Ifter Mahfil CSE 2025	asdfsdf	Tue Mar 11 2025	
8	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	

7.5.6 Interface of a student for Pre-Practicum Offering

General Student Info

Student Name: **Shabit**

Id: **21103098**

Phone No: **01704670152**

Email: **shabitalahi123@gmail.com**

CGPA: **3.38**

Credit Completed: **132**

Practicum Application

Accepted

Your Practicum application is Accepted.

Practicum Supervisor: ASM Shakil Ahmed.

Contact: 01835227954

AO Remarks

Accepted

Last Updated: 3/11/2025, 12:35:56 PM

7.5.7 Interface of a student for Pre-Thesis Offering

Shabit

shabitalahi123@gmail.com

General Requests

Course Add Request

Thesis Request

Practicum Meetings

All Practicum Logs

Add Practicum Log

Pre Course Offering

General Student Info

Student Name: Shabit

Phone No: 01704670152

CGPA: 3.38

Check for Enrollment

Id: 21103098

Email: shabitalahi123@gmail.com

Credit Completed: 132

Thesis Application

Attach your application *

Choose File

Shabit-A-Alahi-Resume (1).pdf

Only one PDF attachment.

Description *

I Want take thesis in this semester and i completed 132 credit. I attached all the required file above

Submit

7.5.8 Interface of a student for Practicum Meeting

CSE Aut

Shabit

shabitalahi123@gmail.com

General Requests

Course Add Request

Thesis Request

Practicum Meetings

Meeting (Tue Apr 22 2025)

Meeting Topics

1. Design a dashboard for view all notice.

Serial	Creation Date	Meeting Date	Meeting
1	Sat Apr 12 2025	Tue Apr 22 2025	Check

Logout

7.5.9 Interface of a student for Perform Task

Shabit

shabitalahi123@gmail.com

General Requests

Course Add Request

Thesis Request

Practicum Meetings

All Practicum Logs

Add Practicum Log

Pre Course Offering

About us

Contact us

Log Book

General Student Info

Student Name: Shabit

Id: 21103098

Phone No: 01704670152

Email: shabitalahi123@gmail.com

CGPA: 3.38

Credit Completed: 132

Date: Sat Apr 12 2025

Completed Activity	Learned Skills
1. What have you done? Html	1. What have you learnt? Htm
2. What have you done?	2. What have you learnt?

Submit

7.5.10 Interface of a Teacher for View All Notices

 CSE Automation

Logout 

ASM Shakil
Ahmed
sahamed.cse@iubat.edu

My Routine

Thesis Students List

Practicum Students List

All Notice

Serial	Title	Description	Date	Action
1	Meeting with chairman sir	Regarding your practicum defense	Sat Apr 12 2025	
2	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	
3	Practicum Offer	Practicum Offer for Fall 2024 Semester	Tue Mar 11 2025	
4	Practicum Offer	Practicum Offer for Summer 2024 Semester	Tue Mar 11 2025	
6	Meeting with chairman sir	For Practicum Board	Tue Mar 11 2025	
7	Ifter Mahfil CSE 2025	asdfs	Tue Mar 11 2025	
8	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	

7.5.11 Interface of a Teacher for fix meeting with Thesis student

 CSE Automation

Logout 

ASM Shakil
Ahmed
sahamed.cse@iubat.edu

My Routine

Thesis Students List

Practicum Students List

All Thesis Students

Serial	Id	Name	Email	CGPA	Student Remarks	Attachments	Action
1	21303010	Md Rabbi	shahidula699@outlook.com	2	asd	Attachment	LogBook Meeting

About us

Contact us

7.5.12 Interface of a Teacher for view all Thesis Meeting

ASM Shakil
Ahmed
sahamed.cse@iubat.edu

My Routine

Thesis Students List

Practicum Students List

About us Contact us

General Student Info

Student Name: Md Rabbi

Id: 21303010

Phone No: 01835227954

Email: shahidula699@outlook.com

CGPA: 2

Credit Completed: 131

Schedule New Meeting

Previous Meetings

Serial	Creation Date	Meeting Date	Meeting
1	Tue Mar 11 2025	Wed Mar 12 2025	Check
2	Tue Mar 11 2025	Tue Mar 11 2025	Check
3	Tue Mar 11 2025	Wed Mar 05 2025	Check

7.5.13 Interface of a Teacher for fix Meeting and View LogBook of Practicum students

 CSE Automation

Logout [→]

ASM Shakil
Ahmed
sahamed.cse@iubat.edu

My Routine

Thesis Students List

Practicum Students List

About us Contact us

All Practicum Students

Serial	Id	Name	Email	CGPA	Office Address	Attachments	Action
1	21303010	Md Rabbi	shahidula699@outlook.com	2	Mohakhali	See Practicum Application See Placement Letter	LogBook Meeting
2	21103098	Shabit	shabitalahi123@gmail.com	3.38	IUBAT,Uttara	See Practicum Application See Placement Letter	LogBook Meeting

ASM Shakil Ahmed
sahamed.cse@iubat.edu

My Routine
Thesis Students List
Practicum Students List

Schedule New Meeting

Date *
04/18/2025 10:00 AM

Meeting Topics

1. Learn about micro service architecture
2.

Schedule

2	Tue Mar 11 2025	Tue Mar 11 2025	Check
3	Tue Mar 11 2025	Wed Mar 05 2025	Check

ASM Shakil Ahmed
sahamed.cse@iubat.edu

My Routine
Thesis Students List
Practicum Students List

Log Book

Date: Sat Mar 08 2025
Status: Checked

Completed Activity	Learned Skills
1. Asd	1. asd
2. asd	

Date: Sun Mar 09 2025
Status: Checked

Completed Activity	Learned Skills
1. dsdffa	1. asdffa

Date: Mon Mar 10 2025
Status: Checked

CSE Automation

Logout

ASM Shakil Ahmed
sahamed.cse@iubat.edu

My Routine
Thesis Students List
Practicum Students List

General Student Info

Student Name: **Shabit**
Phone No: **01704670152**
CGPA: **3.38**

Id: **21103098**
Email: **shabitalahi123@gmail.com**
Credit Completed: **132**

Schedule New Meeting

Previous Meetings

Serial	Creation Date	Meeting Date	Meeting
1	Sat Apr 12 2025	Tue Apr 22 2025	Check

[About us](#)
[Contact us](#)

7.5.14 Interface for add Semesters

Teachers

↑

Add Teacher

All Teachers

Semesters

↑

Add Semester

All Semesters

Courses

↓

Time Slots

↓

Room Allocation

↓

Generate Section

↓

Routine

↓

Add Semester Information

Semester Name *

Enter Semester Name. Ex: Summer 2024

Year *

Enter Year

Select Semester *

Select...

▼

Start Date *

mm/dd/yyyy

End Date *

mm/dd/yyyy

Submit

7.5.15 Interface for view all Semester



Routine Master

Routine Master

Teachers ↑

Add Teacher

All Teachers

Semesters ↑

Add Semester

All Semesters

Courses ↓

Time Slots ↓

Room Allocation ↓

Generate Section ↓

Routine ↓

View All Semesters

Serial	Semester Name	Year	Start Date	End Date	Action
1	Spring 2024	2024	Sat Jan 06 2024	Mon Apr 01 2024	

7.5.16 Interface for add Teacher information

 Routine Master

Teachers ↑

Add Teacher

All Teachers

Semesters ↓

Courses ↓

Time Slots ↓

Room Allocation ↓

Generate Section ↓

Routine ↓

Add Teacher Information

Name

Enter Teacher Name

Short Code

Enter Teacher Short Code

Email

Enter Teacher Email

Phone Number

Enter Teacher Phone Number

Teacher Information

Choose File

No file chosen

Upload an CSV file with fields (Serial, Name, Short code, Phone, Email)

Submit

7.5.17 Interface for view all Teacher

 Routine Master

Teachers ↑

Add Teacher

All Teachers

Semesters ↓

Courses ↓

Time Slots ↓

Room Allocation ↓

Generate Section ↓

Routine ↓

View All Teachers

Serial	Teacher Name	Teacher ShortCode	Teacher Email	Teacher Phone	Action
1	Rashedul Islam	RI	rashed@iubat.edu	1758975671	 
2	Shahinur Alam	SA	shaheen@iubat.edu	1758975672	 
3	Prof Dr Ehteshamul Haque	EH	ehaquebd@yahoo.com	1758975673	 
4	Md Alamgir Bhuyan	MAB	alamgir@iubat.edu	1758975674	 
5	Dr Hasibur Rashid Chayon	HRC	chayon@iubat.edu	1758975675	 
6	Krishna Das	KD	krishna.das@iubat.edu	1758975678	 
7	Saidur Rahman	SR	s.rahman@iubat.edu	1758975679	 
8	Arifa Tur Rahman	AR	arifarahman@iubat.edu	1758975680	 
9	Md Alomgir Hossain	MAH	alomgir.hossain@iubat.edu	1758975681	 
10	Rubayea Ferdows	RF	rubayea@iubat.edu	1758975676	 

7.5.17 Interface for Room Allocation

 Routine Master

Routine Master

Teachers

Semesters

Courses

Time Slots

Room Allocation

New Allocation

All Records

Generate Section

Routine

Room Allocation

Select Semester *

Select...

Select Time Slot *

Select...

Total Rooms *

0

Submit

7.5.18 Interface for view all Rooms

 Routine Master

Routine Master

Teachers

Semesters

Courses

Time Slots

Room Allocation

New Allocation

All Records

Generate Section

Routine

Room Allocations

Serial	Semester	Time Slot	Total Rooms	Booked Rooms	Available Rooms	Actions
1	Spring 2024	8.30 - 9.30	5	1	4	
2	Spring 2024	9.45 - 10.45	10	3	7	
3	Spring 2024	1.20 - 2.20	5	3	2	
4	Spring 2024	4.20 - 5.20	6	1	5	

7.5.19 Interface for add Course information

Routine Master

Routine Master

- Teachers ↑
- Add Teacher
- All Teachers
- Semesters ↑
- Add Semester
- All Semesters
- Courses ↑
- Add Course
- All Courses
- Time Slots ↓
- Room Allocation ↓
- Generate Section ↓
- Routine ↓

Add Course Information

Course Name *

Course Code *

Credit Hours *

Course Information

No file chosen

Upload an CSV file with fields (Serial, Course Name, Course Code, Credit Hours)

Submit

7.5.20 Interface for view all Course



Routine Master

Routine Master

Teachers ↑

Add Teacher

All Teachers

Semesters ↑

Add Semester

All Semesters

Courses ↑

Add Course

All Courses

Time Slots ↓

Room Allocation ↓

Generate Section ↓

Routine ↓

View All Courses

Serial	Course Name	Course Code	Credit Hours	Action
1	Programming in C	CSC 183	3	✎ 🗑
2	Programming in C Lab	CSC 184	1	✎ 🗑
3	Software Engineering	CSC 469	3	✎ 🗑
4	Software Engineering Lab	CSC 470	1	✎ 🗑

7.5.21 Interface for Generate Section

Ce

Routine Master

Teachers

↓

Semesters

↓

Courses

↓

Time Slots

↓

Room Allocation

↓

Generate Section

↑

Generate Section

All Sections

Assign Teacher

Routine

↓

Generate Section for a Semester

Select Semester *
Spring 2024

Select Course *
Software Engineering (CSC 469)

Total Students *
250

Students Per Section *
50

☐ Create a single section only

Next

7.5.22 Interface for view all Sections

Ce

Routine Master

Teachers

↓

Semesters

↓

Courses

↓

Time Slots

↓

Room Allocation

↓

Generate Section

↑

Generate Section

All Sections

Assign Teacher

Routine

↓

Filter by Semester
Select...

Filter by Course
Select...

View All Sections

Serial	Semester	Course Name	Course Code	Section Name	Student Count	Action
1	Spring 2024	Programming in C	CSC 183	A	50	 
2	Spring 2024	Programming in C	CSC 183	B	50	 
3	Spring 2024	Programming in C	CSC 183	C	50	 
4	Spring 2024	Programming in C	CSC 183	D	50	 
5	Spring 2024	Programming in C	CSC 183	E	50	 
6	Spring 2024	Programming in C	CSC 183	F	50	 
7	Spring 2024	Programming in C	CSC 183	G	50	 
8	Spring 2024	Programming in C	CSC 183	H	50	 
9	Spring 2024	Programming in C	CSC 183	I	50	 
10	Spring 2024	Programming in C	CSC 183	J	50	 
11	Spring 2024	Programming in C Lab	CSC 184	A	45	 

localhost:5173/section/all

Content Script

7.5.23 Interface for Generate Routine



Routine Master

Teachers

Semesters

Courses

Time Slots

Room Allocation

Generate Section

Routine

Generate Routine

Get Teacher Routine

Get Course Routine

Select Teacher

Rashedul Islam (RI)

Select Semester

Spring 2024

Select Section

Select...

Select Day

Select...

Select Time Slot

Select...

Add To Routine

Routine

Day	8.30 - 9.30	9.45 - 10.45	1.20 - 2.20	4.20 - 5.20
Saturday	CSC 183 - C	CSC 183 - C	CSC 183 - I	
Sunday		CSC 183 - C	CSC 183 - C	
Monday			CSC 183 - F	
Tuesday				
Wednesday				CSC 183 - A
Thursday		CSC 183 - I		
Friday				

7.5.24 Interface for view Course Routine



Routine Master

Teachers

Semesters

Courses

Time Slots

Room Allocation

Generate Section

Routine

Generate Routine

Get Teacher Routine

Get Course Routine

Select Course

Programming in C (CSC 183)

Select Semester

Spring 2024

Get Routine

Routine

Day	8.30 - 9.30	9.45 - 10.45	1.20 - 2.20	4.20 - 5.20
Saturday	CSC 183 - C	CSC 183 - C	CSC 183 - I	
Sunday		CSC 183 - C	CSC 183 - C	
Monday			CSC 183 - F	
Tuesday				
Wednesday				CSC 183 - A
Thursday		CSC 183 - I		
Friday				

Print Routine

© 2024

Content Security

7.5.25 Interface for Assign Course to Teacher

Cse

Routine Master

Teachers

Semesters

Courses

Time Slots

Room Allocation

Generate Section

Generate Section

All Sections

Assign Teacher

Routine

Filter by Semester

Filter by Course

Assign Teacher to Section

Serial	Semester	Course Name	Course Code	Section Name	Teacher	Action
1	Spring 2024	Programming in C	CSC 183	A	Rashedul Islam (RI)	
2	Spring 2024	Programming in C	CSC 183	B	Shahinur Alam (SA)	
3	Spring 2024	Programming in C	CSC 183	C	Rashedul Islam (RI)	
4	Spring 2024	Programming in C	CSC 183	D	Rashedul Islam (RI)	
5	Spring 2024	Programming in C	CSC 183	E	Rashedul Islam (RI)	
6	Spring 2024	Programming in C	CSC 183	F	Rashedul Islam (RI)	
7	Spring 2024	Programming in C	CSC 183	G	Rashedul Islam (RI)	
8	Spring 2024	Programming in C	CSC 183	H	Rashedul Islam (RI)	
9	Spring 2024	Programming in C	CSC 183	I	Rashedul Islam (RI)	
10	Spring 2024	Programming in C	CSC 183	J	Rashedul Islam (RI)	

Content Script

7.5.26 Interface for view Teacher Routine

Cse

Routine Master

Teachers

Semesters

Courses

Time Slots

Room Allocation

Generate Section

Routine

Generate Routine

Get Teacher Routine

Get Course Routine

Select Teacher

Select Semester

Get Routine

Routine

Day	8.30 - 9.30	9.45 - 10.45	1.20 - 2.20	4.20 - 5.20
Saturday	CSC 183 - C	CSC 183 - C	CSC 183 - I	
Sunday		CSC 183 - C	CSC 183 - C	
Monday			CSC 183 - F	
Tuesday				
Wednesday				CSC 183 - A
Thursday		CSC 183 - I		
Friday				

Print Routine

Content Script

7.5.27 Interface for view TimeSlot



Routine Master

Routine Master

Teachers ↑

Add Teacher

All Teachers

Semesters ↑

Add Semester

All Semesters

Courses ↑

Add Course

All Courses

Time Slots ↑

Add Time slot

All Time slots

Room Allocation ↓

Generate Section ↓

Routine ↓

View All Time Slots

Serial	Time Slot	Alternative Time Slot	Action
1	8.30 - 9.30	8.00 - 9.00	 
2	9.45 - 10.45	9.25 - 10.25	 
3	1.20 - 2.20	1.00 - 2.00	 
4	4.20 - 5.20	4.00 - 5.00	 

Content Script

7.5.28 Interface of view all Notices

 CSE Automation

Logout 

Utpal Kanti Das

ukd.admin@iubat.edu

Practicum 

Thesis 

All Notice

Serial	Title	Description	Date	Action
1	Thesis Offer	Thesis Offer for Spring 2024 Semester	Sat Apr 12 2025	  
2	Practicum Offer	Practicum Offer for Spring 2024 Semester	Tue Mar 11 2025	  
3	Practicum Offer	Practicum Offer for Fall 2024 Semester	Tue Mar 11 2025	  
4	Practicum Offer	Practicum Offer for Summer 2024 Semester	Tue Mar 11 2025	  
5	Meeting with Chairman sir	Regarding your practicum defense	Tue Mar 11 2025	 
6	Meeting with chairman sir	For Practicum Board	Tue Mar 11 2025	 
7	Ifter Mahfil CSE 2025	asdfs	Tue Mar 11 2025	 

7.5.29 Interface of Check Practicum Students Status

 CSE Automation

Logout 

Utpal Kanti Das

ukd.admin@iubat.edu

Practicum 

Practicum Students

Thesis 

Practicum Students

Filter by Semester

Summer 2024 

Serial	Id	Name	Email	CGPA	Assigned Faculty	Action
1	21103098	Shabit	shabitalahi123@gmail.com	3.38	ASM Shakil Ahmed	Log Book

 CSE Automation

Logout 

Utpal Kanti Das

ukd.admin@iubat.edu

Practicum 

Practicum Students

Thesis 

Log Book

Date: Sat Apr 12 2025

Status: Checked

Completed Activity	Learned Skills
1. html	1. html

7.5.30 Interface of Check Thesis Students status

 CSE Automation Logout [->]

Utpal Kanti Das
ukd.admin@iubat.edu

Practicum ↑

Practicum Students

Thesis ↑

Thesis Students

Thesis Students

Filter by Semester

Fall 2024

Serial	Id	Name	Email	CGPA	Assigned Faculty	Action
1	21303010	Md Rabbi	shahidula699@outlook.com	2	ASM Shakil Ahmed	<button>Log Book</button>

 CSE Automation Logout [->]

Utpal Kanti Das
ukd.admin@iubat.edu

Practicum ↑

Practicum Students

Thesis ↑

Thesis Students

Log Book

Date: Tue Mar 11 2025

Status: Checked

Completed Activity	Learned Skills
1. ads	1. asd

Date: Wed Mar 12 2025

Status:

Not Filled

Date: Thu Mar 13 2025

Status:

Not Filled

7.6 Database Design

7.6.1 Normalization of Data Fields

7.6.1.1 What is Database Normalization?

Database normalization is a systematic process of organizing the columns and tables of a relational database to minimize data redundancy and improve data integrity. It involves dividing large tables into smaller, more manageable tables and defining relationships between them. The goal is to isolate data so that additions, deletions, and modifications (updates) of a field can be made in just one table and then propagated through the rest of the database via the defined relationships.

Normalization helps to:

- **Reduce Data Redundancy:** Avoids storing the same information multiple times.
- **Improve Data Integrity:** Ensures data is consistent and accurate across the database.
- **Enhance Data Consistency:** Reduces the chances of conflicting data.
- **Simplify Data Maintenance:** Makes it easier to insert, update, and delete data.
- **Optimize Query Performance (in some cases):** While joining tables can add overhead, well-normalized tables often lead to smaller table sizes and more efficient indexing, which can improve query performance for complex operations.

Normal Form	Rule Description	Purpose
1NF	Atomic columns, unique rows	Eliminate repeating groups
2NF	Full functional dependency on primary key	Eliminate partial dependencies
3NF	No transitive dependency on primary key	Eliminate transitive dependencies

Table 7.6.1: Summary Table of Normalization Stages

7.6.1.2 Normalization Steps

First Normal Form (1NF): Ensure all attributes are atomic (no repeating groups or multi-values).

Second Normal Form (2NF): Remove partial dependencies (i.e., attributes that depend only on part of a composite key).

Third Normal Form (3NF): Remove transitive dependencies.

7.6.2 Database table structure

The screenshot displays the 'dbo.Admins' table structure in SQL Server Enterprise Manager. The table has the following columns:

Column Name	Data Type	Allow Nulls
Id	uniqueidentifier	☐
Name	nvarchar(MAX)	☐
Email	nvarchar(MAX)	☐
Phone	nvarchar(MAX)	☐
PhoneNumber	nvarchar(MAX)	☐
Address	nvarchar(MAX)	☐
Role	nvarchar(MAX)	☐
CreatedBy	uniqueidentifier	☐
CreatedAt	datetime2(7)	☐
CreatedBy	uniqueidentifier	☐
UpdatedAt	datetime2(7)	☒
UpdatedBy	uniqueidentifier	☒
IsActive	bit	☐
EventId	uniqueidentifier	☒
IpAddress	nvarchar(MAX)	☒
MacAddress	nvarchar(MAX)	☒

The 'Column Properties' pane shows the following details for the 'Id' column:

- (General)
 - (Name) Id
 - Allow Nulls No
 - Data Type uniqueidentifier
 - Default Value or Binding (newid())
- Table Designer (General)

Table 7.6.1: Database Table Structure

The screenshot displays the 'dbo.StudentInfo' table structure in SQL Server Enterprise Manager. The table has the following columns:

Column Name	Data Type	Allow Nulls
Id	uniqueidentifier	☐
StudentId	nvarchar(8)	☐
BatchId	uniqueidentifier	☒
Name	nvarchar(MAX)	☐
CGPA	real	☐
ImageUri	nvarchar(MAX)	☒
Address	nvarchar(MAX)	☒
Email	nvarchar(MAX)	☒
IsRegistered	bit	☐
CreditCompleted	int	☐
ContactNo	nvarchar(11)	☐
GradeId	uniqueidentifier	☒
ProgramId	uniqueidentifier	☒
CreatedBy	uniqueidentifier	☐
CreatedAt	datetime2(7)	☐
CreatedBy	nvarchar(255)	☐
UpdatedBy	uniqueidentifier	☒
UpdatedAt	datetime2(7)	☒
UpdatedBy	nvarchar(255)	☒
IsActive	bit	☐

The 'Column Properties' pane shows the following details for the 'Id' column:

- (General)
 - (Name) Id
 - Allow Nulls No
 - Data Type uniqueidentifier
 - Default Value or Binding (newid())
- Table Designer (General)

Table 7.6.2: Database Table Structure

Chapter 8

System Quality & Testing

8.1 System Quality Management

Organizations following rules or meeting ISQ standards need a computerized quality management software that works smoothly across all departments. This software, known as a Quality Management System (QMS) or Total Quality Management (TQM) system, links every stage of making a product and different parts of the company. This creates an environment where everyone can share helpful ideas. Automated routing quickly sends information to the right departments using escalation methods.

This way of doing things shifts the focus from just checking quality (QA) to integrating quality throughout making a product, helping teams in engineering, regulatory work, QA, making things, sales, and marketing.

When checking software quality, we consider different factors, which fall into two types: external and internal quality standards. External quality is about what users experience when they use the software. On the other hand, internal quality is about things in the program's code that users can't see. External quality matters to users, while internal quality is more about what developers care about.

There are mainly two types of quality:

1. Internal Quality
2. External Quality

Internal quality:

- Test coverage
- Testability
- Portability
- Thread-safeness
- Conciseness
- Maintainability
- Scalability

External quality:

- Features
- Speed
- Space
- Network usage
- Stability
- Robustness
- Ease-of-use

8.1.1 Software Quality Management Process**Quality assurance process:**

The goal of quality assurance is to ensure that a project is finished without problems and in accordance with all agreed-upon standards, specifications, and functional requirements. To do this, efforts are made to enhance the development process from the very beginning of the project and to monitor it continuously. There is a strong focus on "prevention."

Verification & validation process:

As part of the verification process, the customer can either go over the code or see it run in a simulated setting. The customer now has very certain software requirements and expectations, thus it is critical that they check that the final product contains all the features that were originally specified. This guarantee is necessary before moving on to the next round of verification and validation. As one of the last steps before a project is finished, this phase isn't the last chance to make changes to the program so it can work as intended. Do not rush to accept the program at this stage as it may cause problems down the road, leading to more expenses for software revisions.

An impartial assessment of how well software and related processes adhere to all applicable regulations, policies, guidelines, plans, and procedures is what software audits are all about. Individuals taking part in this audit are expected to adhere to certain protocols and duties, including those of lead auditor, auditor, recorder, initiator, and representative of the audited

organization. The purpose of the audit is to find areas of nonconformance and write up a report that the team can use to fix them.

8.2 System Testing

Black box testing:

You don't need to know the code or have an in-depth knowledge of the internal architecture to pass this test. The primary motivators are needs, requirements, and technical details.

White box testing:

A deep familiarity with the code and fundamental design is required to conduct these tests. Code statements and styles are tested individually.

Unit testing:

Testing on the smallest possible scale to verify individual code modules or functions. It is typically the programmer, not the tester, who does this since it requires in-depth knowledge of the code and internal software architecture. Without a well-planned architecture and clean code, it could be required to build test driver modules or test harnesses.

Incremental integration testing:

Maintaining a consistent testing schedule for software updates that include new features. In order for the programmer to finish or test drivers to be created as needed, separate parts of the application's operation need to be independent enough to run smoothly. Responsible for this task are programmers or testers.

Functional testing:

Black box testing is a great way to make sure a program works as expected. Although this is typically the responsibility of software testers, developers should also test their code thoroughly before making it public.

System testing:

Blind testing that doesn't take context into account and relies on detailed requirements documents. A system's constituent parts are collectively referred to by this phrase.

End-to-end testing:

System testing is analogous to how it operates. It comprises testing an entire application environment in a real-life scenario. Connecting to a database, exchanging data via a network, or interacting with other programs or systems could be required.

Load / stress / performance testing:

Using heavy weights to test an application. To identify when a system begins to malfunction or cause issues, it is common practice to simulate a high-traffic scenario in a phone or data network, or on a website.

Usability testing:

To determine the program's ease of use, it is currently undergoing testing. Who decides? The customer or user. A variety of methods can be employed, such as user interviews, surveys, video recordings of user sessions, and others. Usability testing is not a job for programmers or testers, in most instances.

8.3 Test Case Details

Table 8.1: Test Cases

Test Case ID	Test Designed by	Test Priority	Test Design Date	Module Name	Test Executed by	Test Title	Test Execution Date	Description
Test1	Kazi Johir Raihan Suny	Medium	10.05.24	User login screen	Kazi Johir Raihan Suny	Verify Login with valid username and password	10.05.24	Test the user login system
Test2	Kazi Johir Raihan Suny	Medium	10.05.24	User Register screen	Kazi Johir Raihan Suny	Verify with valid email, username, and password	10.05.24	Test the registration system

Test3	Kazi Johir Raihan Suny	High	10.05.24	Pre-course offering	Kazi Johir Raihan Suny	Verify pre- course offering functionality	10.05.24	Test the pre-course offering system
Test4	Kazi Johir Raihan Suny	High	10.05.24	Routine generation	Kazi Johir Raihan Suny	Verify routine generation functionality	10.05.24	Test the routine generation system
Test5	Kazi Johir Raihan Suny	Medium	10.05.24	General requests handling	Kazi Johir Raihan Suny	Verify handling of general requests	10.05.24	Test the general requests handling
Test6	Kazi Johir Raihan Suny	High	10.05.24	Thesis and practicum requests	Kazi Johir Raihan Suny	Verify thesis and practicum requests	10.05.24	Test the thesis and practicum requests system

8.3.1 Test Steps for Test 1

Table 8.2: Test Steps for Test 1

Test Steps	Test Data	Expected Result	Actual Result	Status	Notes
1. Navigate to login page	User= Admin Pass= 12345	Should be able to log in	User is navigated to home page with successful login	Pass	

8.3.2 Test Steps for Test 2

Table 8.3: Test Steps for Test 2

Test Steps	Test Data	Expected Result	Actual Result	Status	Notes
1. Navigate to registration page	Name=Shanto Email=web.shahidul.alam@gmail.com Pass=1234 Phone=01766409973 Address=Uttara	Should be able to register	User is navigated to home page successful registration	Pass	Should follow valid email address. Does not take if it breaks email format

8.3.3 Test Steps for Test 3

Table 8.4: Test Steps for Test 3

Test Steps	Test Data	Expected Result	Actual Result	Status	Notes
1. Navigate to pre-course offering page	Course details	Should be able to offer a course	Course offering created successfully	Pass	

8.3.4 Test Steps for Test 4

Table 8.5: Test Steps for Test 4

Test Steps	Test Data	Expected Result	Actual Result	Status	Notes
1. Navigate to routine generation page	Course schedules, faculty availability	Should be able to generate routine	Routine is successfully generated	Pass	

8.3.5 Test Steps for Test 5

Table 8.6: Test Steps for Test 5

Test Steps	Test Data	Expected Result	Actual Result	Status	Notes
1. Navigate to general requests page	General request details	Should be able to handle request	Request is successfully handled	Pass	

8.3.6 Test Steps for Test 6

Table 8.7: Test Steps for Test 6

Test Steps	Test Data	Expected Result	Actual Result	Status	Notes
1. Navigate to thesis request page	Thesis request details	Should be able to handle request	Thesis request is successfully handled	Pass	

8.3.7 Test Steps for Test 7

Table 8.8: Test Steps for Test 7

Test Steps	Test Data	Expected Result	Actual Result	Status	Notes
1. Navigate to practicum request page	Practicum request details	Should be able to handle request	Practicum request is successfully handled	Pass	

Chapter 9

Ethical Consideration

9.1 Ethical Considerations in the Software Development Process

The development of the Automated Administrative Services Management System was guided by strong ethical principles to ensure the platform promotes transparency, fairness, and reliability. The following ethical aspects were carefully considered throughout the system design and implementation phases:

9.1.1 Data Privacy and Security

This system handles sensitive academic and personal data of students, faculty, and staff. To uphold user privacy and system integrity, several security measures were implemented, including secure password hashing, HTTPS protocols, input validation, and role-based access control. All user data is stored securely, and access is strictly limited to authorized personnel. Activity logging mechanisms are in place to detect and prevent misuse or unauthorized access.

9.1.2 Intellectual Property

All components used in this system, including open-source tools and libraries, comply with their respective licenses. The custom code and original design elements were developed independently, with appropriate citations provided for any third-party content. No proprietary software was used without proper authorization.

9.1.3 User Impact

The system was designed with a student- and faculty-centered approach. Features such as course management, notice handling, and routine generation were built to improve user experience, reduce administrative workload, and ensure accurate information dissemination. Extra care was taken to avoid any system behavior that could lead to academic or administrative errors.

9.1.4 Bias and Fairness

The system ensures equitable treatment of all users, regardless of their academic level or role. Functionalities such as request handling, supervisor assignments, and routine allocations are implemented based on objective, pre-defined criteria to avoid any form of favoritism or discrimination.

9.1.5 Responsibility to Stakeholders

All primary stakeholders including students, faculty, and administrative officers were considered during development. Feedback gathered from users was incorporated into system improvements. The system prioritizes features that have direct benefits for academic operations and departmental transparency

9.1.6 Professional Responsibility

The project adhered to industry best practices and software engineering standards. This included structured documentation, version control, rigorous testing, and adherence to ethical guidelines. Decisions throughout the development lifecycle were made with accountability and quality in mind.

9.2 Sustainability in the Software Development Process

Beyond technical functionality, sustainability was a core consideration in the development of the Automated Administrative Services Management System. The system was designed to be economically viable and socially beneficial over the long term.

9.2.1 Economical Sustainability

To ensure cost efficiency, the system was developed using free and open-source technologies such as React.js, .NET Core Web API, MySQL, and Visual Studio Code. This makes it affordable for educational institutions like IUBAT to implement and maintain without incurring significant overhead costs. The modular design supports future scalability and reduces the need for major rework during system upgrades.

9.2.2 Social Sustainability

The system contributes to academic transparency, student satisfaction, and operational efficiency—all of which enhance the educational experience. By automating routine administrative tasks, it reduces manual errors and ensures timely service delivery. This promotes a fair and responsive environment for students and faculty, ultimately supporting long-term academic and institutional development.

Chapter 10

Conclusion

10.1 Practicum and Its Value

In jobs and life, hard work usually pays off. Doing a practicum, which is like hands-on training in engineering, helps connect what you learn in school to real jobs. There are three big benefits:

- **Show How You Use What You Learn:** You can show how the stuff you learn in school is useful in real life.
- **Get Really Good at Something:** It helps you become an expert in the area you're interested in.
- **Get More Job Opportunities:** It makes it more likely for you to get a job because you've got practical experience.

Practicum not only gives you hands-on experience but also helps you fit into society. Meeting new people and facing new challenges gives you a practical outlook on life. There are challenges in practicum that only the person going through them can truly understand. It's like a bridge between what you learn in school and what you need to know to work in a job.

These days, employers hiring engineers prefer real-world experience over just good grades. Doing a practicum is an advantage for new engineers entering the job market. At the College of Engineering and Technology (CEAT) at IUBAT, students take part in a practicum program that is worth 6 credit hours. It lasts for a semester, usually after finishing regular classes. After finishing the practicum, students write a report, do a presentation, and take an exam covering their entire four years of education.

10.2 Conclusion

I have had the opportunity to contribute to the development and implementation of software as an intern at CSE Department. The design section was where I learned the most. Thanks to this experience, I learned a lot of new things and gained technical abilities that will help me in the future. Some of the technical aspects of this project that I have learned and worked on are listed below:

- A web-based project's technique for design.
- A web-based project's analysis approach.

- A fresh setting for computer languages.
- Making use of the current project to build a new

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